TS5000 Robot Controller

TS5000-SS

Controller Manual

(Compatible Robots: THL500/600/700，THE400)

Rev. 2

October 1, 2022

Important

- Make sure that this instruction manual is delivered to the final user of Shibaura Machine's industrial robot.
- Be sure to read this manual and that you fully understand its contents before operating the industrial robot.
- After reading this manual, keep it in an easily accessible place for future reference.

Shibaura Machine

This instruction manual is intended for the TS5000-SS controller.

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Introduction

Thank you for purchasing this Shibaura Machine robot system.
This manual describes the basic specifications of the TS5000-SS robot controller, how to install the controller, how to connect wiring, and how to maintain the controller.
Read this manual before actually unpacking the product.
Before using this manual, read the STEA1235_THL500/THL600/THL700/TS5000-SS Safety Manual or STEA1239_THE400 Safety Manual to ensure that you fully understand the implemented safety measures.

This manual has the following chapters.

Chapter 1: Specifications
This chapter describes the basic specifications of the controller.

Chapters 2 to 5: Installation
These chapters describe transportation, unpacking, carrying, and installation of the controller.
These chapters also describe the wiring between the robot and controller and between the controller and peripheral devices and the key points at initial operation.

Chapters 6 to 8: Maintenance
These chapters describe the maintenance of the controller.

Chapter 9: Disposal
This chapter describes the disposal of the controller.

Manufacturer

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If you have purchased this controller outside of Japan, please contact the company that you purchased it from.

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Scope of Use

The product has been developed for use in a wide range of automation tasks including transport work such as the movement and alignment of workpieces and assembly work such as insertion of parts and screw tightening.
The customer should also use this product for the purpose of automating these tasks and for incorporating into automation equipment.

This product is not intended for use in tasks involving human contact (such as tasks for medical care and nursing care) or to be incorporated in equipment that sustains human life (such as life support devices and incubators), equipment on which people ride (including transportation equipment such as cars and aerospace equipment such as airplanes, and recreational equipment such as roller coasters), and equipment that protects human life or the human body (such as safety devices).
This product is not intended to be incorporated in equipment for controlling large heat quantities (such as nuclear power controllers and incinerator controllers).
This product is prohibited from being used for tasks or incorporated in equipment where a failure or malfunction could likely pose a direct threat to human life or harm to the human body. The customer is responsible for providing any warranties if this product is used in this way.

Usage Procedures

This section describes the usage conditions where the TS5000-SS can be connected. Usage under different conditions could expose the user to unexpected risks. Be sure to always use the controller in an environment that meets the required conditions.

1. Main Specifications of Product: Refer to the robot manual for your robot.

2. Usage Restrictions: Tables 1 to 6 provide the “Restricted Specifications Sheet” indicating the usage restrictions.

Table 1. Restricted Specifications Sheet

1. Purpose and application for using product (usage restrictions)	
1) Intended use	To make the robot perform operations for specific purposes (item transport, assembly, inspection, machining, etc.) (1) Transportation [1] Clear a transportation route to the unpacking site. [2] Use a forklift or other equipment to lift the packing box. [3] Move the packing box to the unpacking site. (2) Unpacking [1] Clear the space required for unpacking. [2] Cut the packing straps and unpack. [3] Cut the packing straps securing the robot. [4] Peel off the tape from the controller packing box, and unpack. [5] Check that the robot and controller are not damaged. [6] Check the accessories and their quantities. - When transferring the robot to the installation site [7] Dispose of, or store and manage the packing materials. - When storing the robot [8] Secure the robot in place in the same way as the state at delivery, and repack. [9] Move the packing box to the storage site. (3) Transfer - Robot [1] Check the orientation of the robot during transfer. [2] Check that the robot fixing jigs are mounted. [3] If using a crane, mount the supplied eyebolts. [4] Check the tightness of the fixing bolts of the robot fixing jig. [5] Use a crane or cart to transfer the robot to the installation site. - Controller [1] Use a cart to transfer the controller to the installation site.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(4) Installation [1] Use bolts to secure the robot to the frame and other parts. [2] Fabricate any needed cables. [3] Remove the robot fixing jigs and eyebolts. [4] Install the controller and connect it to the robot. [5] Press the robot by hand to move it to a location where it is easy to perform work. [6] Mount the end effector. [7] Connect the required external devices. (5) Teaching [1] Move the robot to the desired position or orientation by guiding it with a teach pendant or pushing it by hand. [2] Teach the position where the robot will operate. [3] Create the operation sequence (program, etc.). [4] Make sure that the signals and data are transferred to and from the external equipment. [5] Check operation at low speed (250 mm/sec or less). [6] While gradually increasing the operating speed, make sure that the operation matches the specific purpose. (6) Operation [1] Start robot operation. [2] Make sure that operation is being performed to match the specific purpose at the specified speed. [3] Stop robot operation. (7) Repacking [6] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Obtain the space needed for packing. [6] If using a crane to transfer the robot, mount the supplied eyebolts. [7] Transfer the robot and controller to the packing site. [8] Pack the robot and controller so that they are in the original delivery state. [9] Clear a transportation route to the transportation equipment (truck, etc.). [10] Use a forklift to lift the packing box. [11] Load the packing box onto the transportation equipment.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
1) Intended use (continued)	(8) Maintenance Work performed when the primary power is off: - Robot [1] Check the bolts for any looseness and backlash and re-tighten them. [2] Make sure that the cable clamp is not loose. [3] Check for any damage. [4] Clean off any dust. [5] Inspect and apply corrosion inhibitor. [6] Check for any wear on the cable and air tube. [7] Check that there is no misalignment between the motion of each joint and the origin position. - Controller [1] Replace the memory backup batteries in the controller. Work performed when the primary power is on: [1] Check the greasing state and apply grease if needed (when the servo power is off). [2] Check for any abnormal vibrations and sounds during operation (when the servo power is on). [3] Check the teaching point for any misalignment (when the servo power is on). [4] Replace the backup batteries for the robot position detector (when the servo power is off). (9) Disposal [6] Remove the end effector. [2] Move the robot to the position and orientation where the fixing jig can be mounted by guiding it with a teach pendant or pushing it by hand. [3] Mount the fixing jig. [4] Turn off the power, and remove the wiring. [5] Use a crane to transfer the robot, and mount the supplied eyebolts. [6] Transfer the robot and controller to the disposal site. [7] Dispose of the robot and controller using a method compliant with your country’s laws and regulations.

Table 1. Restricted Specifications Sheet (continued)

1. Purpose and application for using product (usage restrictions)	
2) Foreseeable misuse	- Use for work which applies excessive impact to the robot or where excessive impact is applied by the robot - Modification of robot, controller, and optional parts, or supplying power while these covers are removed - Operation in other than the specified installation state (floor, suspended, and wall-mounted) - Operation on a frame or other structure that does not satisfy the specified strength - Operation under conditions not satisfying the specified installation requirements - Entry into the hazard zone of the robot by mistakenly assuming that a stop to wait for an operation signal during operation was an operation stop - Usage under conditions where the specification values are exceeded or usage under improper settings - Stop of the robot caused by power off during operation or when the servo power is on - Unintended operation caused by pressing a wrong button - Operating without checking operation in MANUAL mode - Operating without checking the tightness of the robot and hand set bolts - Unintended operation due to improper wiring - Unintended operation due to incorrect program selection - Riding on or leaning against the robot or hanging on the arm
3) Unexpected start	- Malfunction due to excessive noise - Robot start due to start signal unexpectedly sent from peripheral equipment to the robot - Unintended start due to abnormal communication data - Malfunction due to voltage fluctuation - Power on due to powering after power outage - Malfunction due to failure or the like

Table 2. Restricted Specifications Sheet

2. Replacement or other modification of program components (time restrictions)	
1) Mechanical restrictions	- Before operating the robot, perform operation at a low speed (20% or less of maximum speed) for 5 minutes - Replace consumable parts as required or periodically by conducting daily inspection, regular inspection (every 3 and 6 months and every year) and overhauls (every 5 years).
2) Electrical restrictions	

Table 3. Restricted Specifications Sheet

3 Movable range of product (restriction on space)		
1) Work envelope	1st axis 2nd axis 3rd axis 4th axis Additional axis — —	Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with 1.4 Specifications Table Compliant with individual specifications. — —
2) Interface	Transportation: Unpacking: Transfer: Installation: Teaching: Operation: Repacking: Maintenance: Disposal: 	The packing box shall be able to be transferred by a transportation worker. The packing box shall be able to be carried in. Space of at least 2 m^3 shall be provided. The robot shall be able to be hoisted by a crane. The robot shall be able to be lifted by two or more people and loaded onto a cart or similar device. There shall be no obstacles on the route from the unpacking site to the installation site. The product shall be installed at a location suitable for the outline and movable range of the robot. The product shall be installed at a location suitable for the outline and that provides adequate ventilation of the controller. The cables shall be installed at a location where they will not be stepped on by operators. Teaching work shall be able to be performed from outside the hazard zone of the robot. Space shall be provided so that, if a person enters the hazard zone, the person will not get entrapped by the robot or peripheral equipment. A fence or enclosure shall be installed to prevent any operator from approaching the hazard zone of the robot. The robot shall be able to be operated from outside the hazard zone of the robot. An operator in the fence or enclosure shall be visible from the operation location. Work shall be able to be performed to remove the robot hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the packing site. The repacking work shall be able to be performed. A forklift or other equipment shall be able to be used to lift the packing box. Space required for maintenance shall be provided. After maintenance, operation shall be able to be checked from outside the hazard zone of the robot. Space shall be provided so that, in the hazard zone, a person will not get entrapped by the robot or peripheral equipment. Work shall be able to be performed to remove the robot hand, mount the fixing jig, separate the robot and controller, and remove the robot and controller from the frame. The robot shall be able to be moved from the installation site to the disposal location.

Table 3. Restricted Specifications Sheet (continued)

3. Movable range of product (restrictions on space)	
3) Work environment	- A fence, enclosure, or other protection shall be provided that surrounds the hazard zone of the robot to prevent people from entering the area during operation. - A display indicating the robot operation status shall be provided to indicate whether the robot is operating or stopped. - Lighting shall be provided for enabling visual checks by operators. - To prevent operators from slipping, the floor shall not be covered with oil or similar substances. - No noises that impair attention shall be generated.

Table 4. Restricted Specifications Sheet

4. Life cycle of product
In the product life cycle of assembly, shipment testing, transportation, unpacking, transfer, installation, teaching, operation, repacking, maintenance, and disposal, the risk assessment is carried out for the life cycle of “transfer and installation” (transportation, unpacking, transfer, installation, and repacking), “use” (teaching, operation, and maintenance), and “disposal”.

Table 5. Restricted Specifications Sheet

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
1) Transportation worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for forklifts and other transfer equipment work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
2) Unpacking worker	Knowledge	Contents of the instruction manual
	Requirement	—
3) Transfer worker	Knowledge	Contents of the instruction manual
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
4) Installation worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
5) Teaching worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	Person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
6) Operator	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations
	Requirement	—
7) Repacking worker	Knowledge	Contents of the instruction manual, operating procedures for target robot
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer

Table 5. Restricted Specifications Sheet (continued)

5. Person who approaches the machine		
Applicable person	Knowledge and requirements	
8) Maintenance worker Work supervisor	Knowledge	Contents of the instruction manual, operating procedures for target robot, work regulations, and robot operations Storage procedures for target robot
	Requirement	Person who has received training for robot inspection work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
9) Disposal worker	Knowledge	Contents of instruction manual, operating procedures for target robot, and waste disposal methods
	Requirement	Person who has received training for crane work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer When guiding a robot, a person who has received training for robot teaching work as defined by the respective country's laws and regulations and who has been permitted to perform the work by the employer
10) People passing nearby Other workers	- Measures shall be taken to prevent entry into the fence, enclosure, or other protection surrounding the hazard zone of the robot. - Measures shall be taken to make the robot stop operation when someone approaches the hazard zone. - If a person enters the hazard zone of the robot to perform work, robot operation shall be stopped, the drive power turned off, and a teaching worker or maintenance worker shall supervise the situation to ensure safety.	

Table 6. Restricted Specifications Sheet

6. Other

Warranty

We conduct strict testing and inspection of this product before delivery to ensure that it satisfies our performance standards.
We warrant this product to be free from defects under the following terms and conditions.

1. Warranty period

The warranty period of the product is either of the following, whichever is earlier.

(1) 18 months after shipping from our factory
(2) 12 months after installation at customer site

2. Warranty scope

(1) The warranty is provided for this product. The warranty covers only the specifications and functions included in the product specifications, catalog, and instruction manual. Any secondary or incidental damages caused by failure of this product are not covered by the warranty under any circumstances.
(2) We will repair a faulty product free of charge only if this product has been operated and used in accordance with the instruction manual supplied with the product.
A fee will be charged for any repairs after the warranty period has expired.

3. Disclaimer

The warranty shall not apply to the following cases.

(1) Any failure or damage arising from careless use or misuse, or usage that is not described in the instruction manual.
(2) Any failure caused by aging or normal wear and tear (natural fading of paints, deterioration of consumable parts[1]).
(3) Irregularities confined to sensory quality such as noise not recognized as affecting functions
(4) When any modification or disassembly without our consent was performed
(5) Any failure or damage arising from inadequate maintenance and inspection or improper repair
(6) Any failure or damage due to an act of God, fire, and other force majeure
(7) Internal data such as programs and points created or modified by customers
(8) When a product purchased in Japan is taken abroad

*1): Consumable parts refer to the replacement parts for maintenance as listed in the manual of the controller.

4. Safety Information

(1) Shibaura Machine shall provide no guarantees for the standard performance of this product when used outside the specifications.
(2) If the customer does not observe the safety information appearing for DANGER, WARNING, CAUTION, and NOTES contained in this instruction manual, Shibaura Machine will not be liable for any resulting injuries, death, or other accidents, damage, product failure, or any other problems.
(3) Note that the safety information appearing for DANGER, WARNING, CAUTION, and other explanations in this instruction manual are within the scope of usage assumed by Shibaura Machine, and they do not foresee all possible incidents for all situations that could occur.

Standards, laws, and regulations pertaining to the safety of industrial robots

The international industrial standard ISO/DIS 12100 "Safety of machinery" specifies risk reduction measures for machinery and equipment.
Step 1: Inherently safe design measures - Limitations on force, speed, energy, etc.
Step 2: Safeguarding - Installation of safety fences, etc.
Complementary protective measures - Installation of emergency stop devices, etc.
Step 3: Information for use - Warning signs, alarms, instruction manuals, etc.
Based on the above, the ISO/IEC has defined "Guide 51", which classifies the various standards in a hierarchal structure.

Safety standards for industrial robots are standardized as separate safety standards for machinery in Type C.

- ISO 10218, ISO 10218-1

Be sure to always follow the standards, laws, and regulations set forth in the country where the industrial robot is used.

Safety Information

This instruction manual contains important information to ensure the safe and correct use of the controller and to prevent injury to the operator and others and damage to property.
To ensure that the procedures from installation to operation of the controller are performed safely, be sure to carefully read the separate instruction manual STEA1235_THL500/THL600/THL700/TS5000-SS Safety Manual or STEA1239_THE400 Safety Manual before starting actual operation.

Be sure that you fully understand the following information (signal words and symbols) before reading the main text, and always follow the provided safety instructions.

Display of Signal Words and Their Meanings

Signal Word and Symbol	Meaning
⚠ DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
⚠ WARNING	Indicates a potentially hazardous situation which, if not avoided, will result in death or serious injury.
⚠ CAUTION	Indicates the possibility that incorrect operation could cause injury[1] to the operator.
NOTICE	Indicates the possibility that incorrect operation could cause property damage[2] only and provides helpful advice about performing work.

1) Injury refers to wounds, burns, electric shocks, and similar cases that do not require hospitalization or prolonged outpatient treatment.
2) Damage to property refers to extensive damage to property or materials.

⚠ DANGER
In order to ensure controller safety from transportation to disposal, be sure to carefully read the safety manual shown in STEA1235_THL500/THL600/THL700/TS5000-SS Safety Manual or STEA1239_THE400 Safety Manual. Performing an operation different from what is described in the manual could result in harm to people.

Service Network

Please refer to our distributor's handouts

Specifications

This chapter describes the basic specifications and names of the parts for the TS5000-SS.

1. Specifications

1.1 Equipment Configuration

The TS5000-SS supports the THL500/600/700 and THE400 SCARA robots. For details on how to connect the 200 V power supply, see "4.1.2 "ACIN" connector for connecting the power cable Figure 4.1-①." For details on how to connect the 24V power supply, see "4.1.4 "EMGSTOP" connector for connecting input/output external power supply and external emergency stop input/output Figure 4.1-⑥."

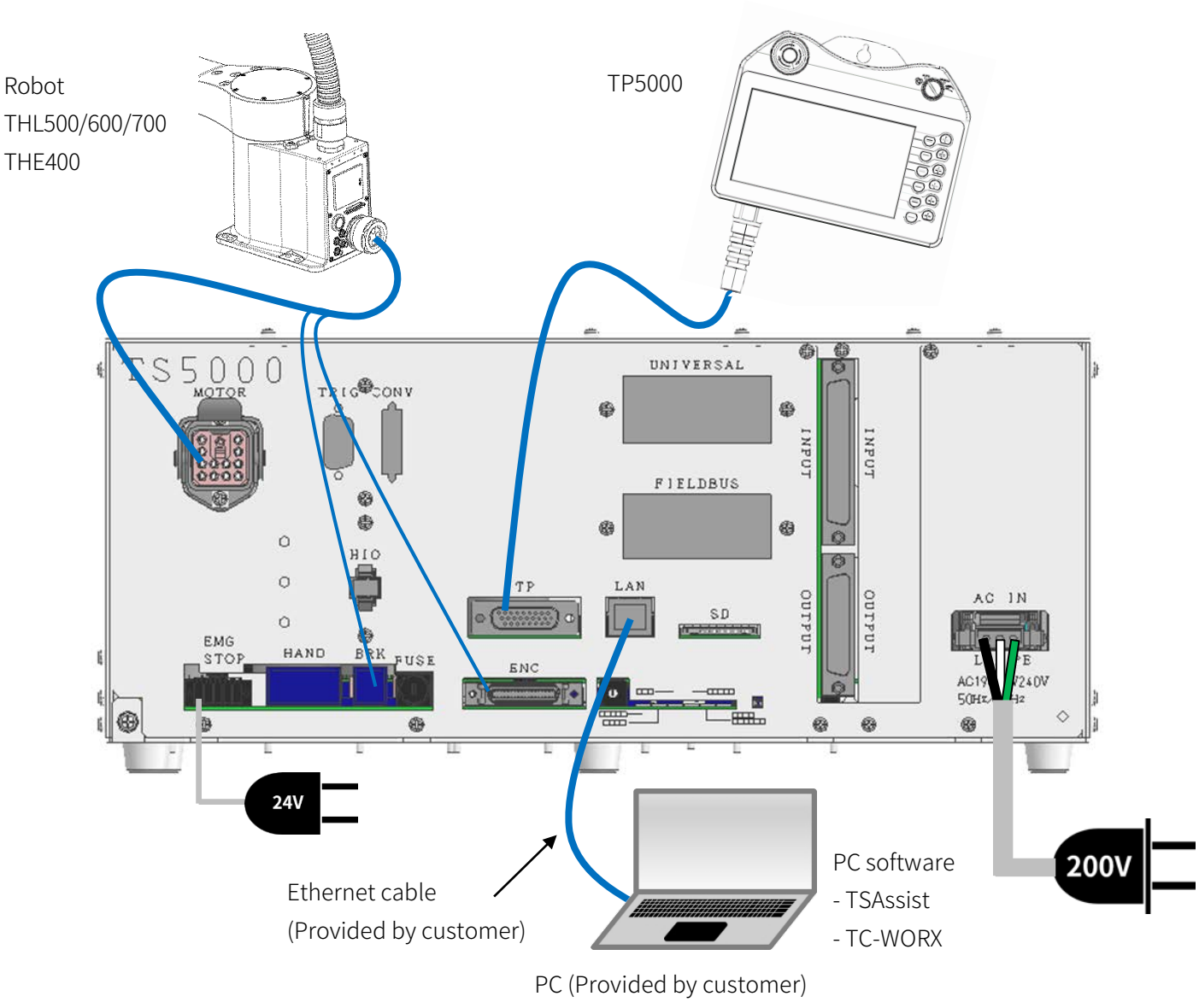


Figure 1.1 Robot Equipment Configuration

1.2 Names of Parts

The connections for the power supply terminal, robot control cable, and EMGSTOP connector are described in "4. Cable Connection".

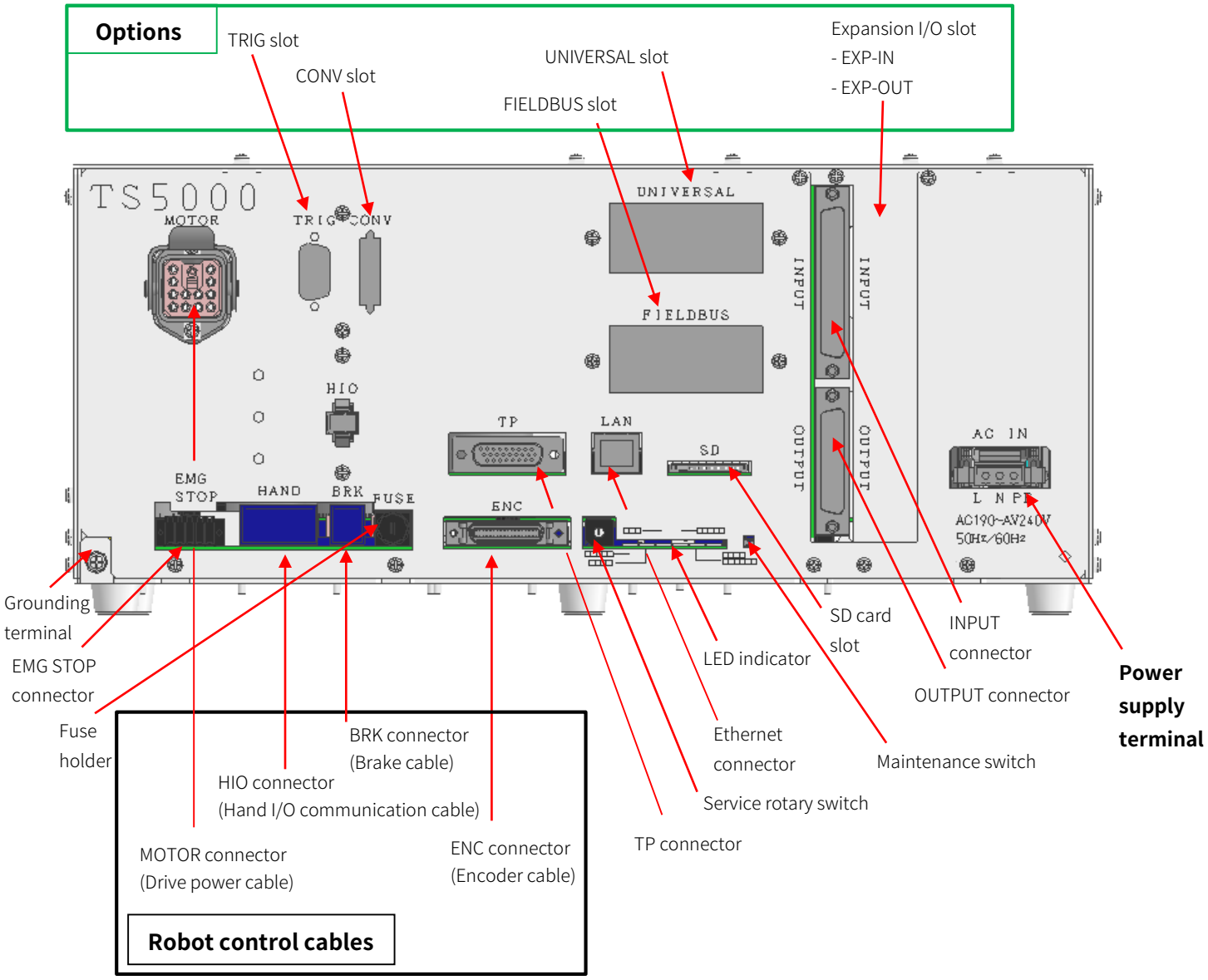


Figure 1.2 Names of Parts

MEMO

1. *The service rotary switch is used by the Shibaura Machine Service Department. This switch should never be touched because touching it carelessly will clear the memory.*

1.3 Controller Standard Specifications

No.	Item		Specifications		Notes
1	Number of controlled axes		4 axes		
2	Servo type		Digital servo		
3	Position detection method		Absolute		
4	Main memory		Controller built-in flash ROM Capacity: 12 MB		Maximum number of writes 10,000
5	Auxiliary memory		SD card (SD, SDHC) Maximum capacity: 32 GB		SDXC is not supported
6	Number of programs that can be stored		Main memory	Up to 512 User files: 502 System files: 10	
			Auxiliary memory	Up to 512 User files: 512	
7	Maximum number of program lines		Per program Data section 5000 points Program section 5000 lines		
8	Programming language		SCOL2		
9	Operation command		PTP, CP (line, arc), shortcut, arch		
10	Teaching device		Teach pendant, Program support software		Options
11	Teaching method	Point teaching	Guidance operation from teaching device		
			Coordinates: Input the X, Y, Z, and C values via the teach pendant.		
			Servo-free: Arm moves by input		
		Program input	Input from teaching device		
12	Master modes		MANUAL mode/AUTO mode		
13	Execution modes		Continuous, cycle, step, step for each operation		
14	Speed setting		Override/Speed limit/Program commands, 1% to 100% for each command		
15	Acceleration setting		Program command: 1% to 100%		
16	Torque control		Program command: 1% to 300%		
17	Coordinate system		Base, workpiece, and tool coordinates can be set.		
18	Movement limit		Soft limit		
19	Self-diagnostic function		Alarm detection, etc.		
20	Interrupt function		Starts an interrupt program via an input signal, timer, etc.		
21	Communication function		Ethernet 1 ch (1 Gbps)		

22	I/O	General-purpose	8 inputs/8 outputs	
		System	13 inputs Program selection, start, stop, reset, etc. 9 outputs Servo On, operation ready, fault, etc.	
		Hand	8 inputs/8 outputs	
		Extension	21 inputs/17 outputs	Options
23	Dimensions		365 (W) × 161 (H) × 325 (D) mm	The height includes the rubber foot height.
24	Weight		11 kg	
25	Power supply		Single-phase, 190 to 240 V AC, 50/60 Hz	
26	Ambient operating temperature		0 to 40°C	
27	Ambient storage temperature		-10 to 50°C	
28	Lifespan parts		Button cell battery (replacement recommended after 5 years) Switching power supply (replacement recommended after 5 years) Large-capacity capacitor (replacement recommended after 9 years)	The actual lifespan will vary depending on the usage conditions.

MEMO

1. *The compatible SD cards are SD and SDHC.*
2. *Use an SD card formatted in FAT16 or FAT32.*

1.4 Specifications for Controller Options

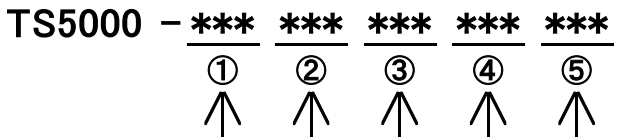
No.	Product Name		Notes
1	Teach pendant	TP5000	Standard cable 5 m
2		TP1000	To connect the TP1000, the "TS5000-TP1000 conversion cable" is required.[1)]
3	TS5000-TP1000 conversion cable		
4	Controller mounting bracket		
5	Extension input/output signal		21 inputs/17 outputs
6	Program support software	TSAssist	Windows application
7	TCmini (built-in PLC) editing software	TC-WORX	Windows application
8	Fieldbus function	CC-Link	2)
9		EtherNet/IP	
10		EtherCAT	
11		PROFINET	
12		DeviceNet	
13		PROFIBUS	
14	Conveyor synchronization function		
15	Vision + Conveyor synchronization function		

1) The TP3000 teach pendant cannot be connected to the TS5000.

2) The Fieldbus slave module type can be selected from CC-Link, EtherNet/IP, EtherCAT, PROFINET, DeviceNet, or PROFIBUS. The Fieldbus to be used must be determined when the order is placed.

CC-Link is a registered trademark of Mitsubishi Electric Corporation.
EtherNet/IP is a registered trademark of ODVA, Inc.
EtherCAT is a registered trademark of Beckhoff Automation GmbH & Co. KG.
PROFINET is a registered trademark of PROFIBUS Nutzerorganisation e.V.
DeviceNet is a registered trademark of ODVA(Open DeviceNet Vendor Association).
PROFIBUS is a registered trademark of PROFIBUS Nutzerorganisation e.V.

1.5 Controller Model Name



Controller type	
SS	Popular type of Controller for SCARA Robot
MS	Standard type of Controller for SCARA Robot
EMS	Controller with CE Specification for SCARA Robot

Conveyor synchronization function	
No symbol	Without conveyor synchronization function (Standard)
–CV	Conveyor 2ch, Trigger input 4ch

FIELDBUS	
No symbol	Without Fieldbus (Standard)
–IP	EtherNet/IP
–PN	PROFINET
–CA	EtherCAT
–CC	CC-Link
–PB	PROFIBUS
–DN	DeviceNet

Extension I/O	
No symbol	Without extension I/O (Standard)
–IO	With extension I/O Addition of general purpose 21 inputs/17outputs

Hand I/O	
–HR	Hand I/O module built in robot (8 inputs/8 outputs)
–HC	Hand I/O module built in controller (8 inputs/8 outputs)/17outputs

Installation

These chapters describe transferring and installation of the controller.

2. Transportation, Unpacking, and Carrying

Transportation, unpacking, and carrying of the controller must be performed by qualified personnel.
These processes must be compliant with the laws, regulations, and ordinances of the respective country.
Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with the robot arm	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with the robot arm	
Protective gloves	Protection part: Hand and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

2.1 Controller Transportation

⚠ CAUTION

- Do not stack the packing boxes. The load could collapse, resulting in an injury.
- Wear protective gloves and shoes when carrying the controller by hand. Dropping could result in crushed hands or feet.
- Use a cart or similar device for transporting the controller. Carrying by hand can lead to dropping, resulting in crushed hands or feet.

NOTICE

- Do not subject the controller to excessive vibrations during transportation. This could damage the controller.
- Do not transport upside down, and do not store on its side. This could damage the controller.
- Do not allow moisture to get inside the controller. This could also lead to a breakdown.

2.2 Unpacking the Controller

The controller is packed and shipped in cardboard boxes. Unpack at a location suitable for later carrying and installation, and unpack while being careful not to damage the robot and controller.
After unpacking, check that nothing has been damaged during transport and check the quantities of the accessories. The controller accessories are shown in 8.3 Accessories List.
After unpacking, dispose of any unneeded wooden pallets, cardboard boxes, plastic bags, cushioning materials, and other items in compliance with the customer's regulations.
To repack, perform the above unpacking procedure in reverse.

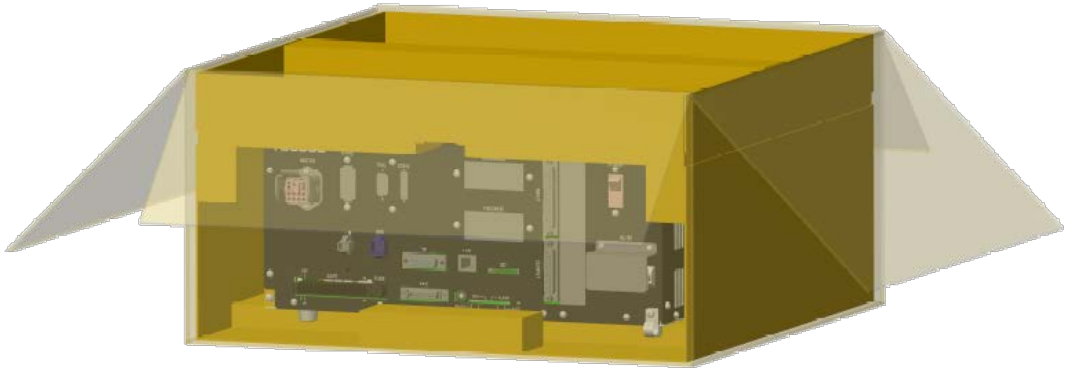


Figure 2.1 Controller Packed State

⚠ WARNING

- When repacking, turn off the power before connecting and disconnecting cables from the controller. Touching the terminals of any cables that were disconnected while power was flowing could result in an electric shock.

⚠ CAUTION

- After unpacking, wear protective gear when disposing of unneeded wooden pallets and cardboard boxes. Otherwise, you could be cut by nails sticking out and by other sharp locations, resulting in an injury.
- Use a box cutter or similar tool when opening the packing box. You could cut your finger or hand if using a cutter with a sharp edge.
- Unpack at a stable location. If unpacked at an unstable location, the controller could fall and crush your hands or feet.
- Wear protective gloves when unpacking. Your hand could get trapped between a cardboard box and the controller, resulting in an injury. You could also be injured by the corners of the controller or cardboard box.

NOTICE

- Do not allow liquids such as water or foreign objects to get inside the controller. This could also lead to a breakdown.

2.3 Carrying of the Controller

⚠ WARNING

- Do not carry the controller while the power is still on. An electric shock will occur if you touch the connectors or terminals while the power is on.

⚠ CAUTION

- Wear protective gloves and shoes when carrying the controller by hand. Dropping could result in crushed hands or feet.
- Use a cart or similar device for transporting and carrying. If carried by hand, the controller could be dropped, resulting in crushed hands or feet.
- Carry with the controller cables removed. Carrying with the cable still connected could cause cable breakage, resulting in an electric shock.

NOTICE

- Do not subject the controller to excessive vibrations. This could damage the controller.
- Do not allow liquids such as water or foreign objects to get inside the controller. This could also lead to a breakdown.

2.4 Safety Information for Storing the Controller

NOTICE

- Store the controller in a location away from direct sunlight. Otherwise, the inside of the controller could become hot, resulting in a breakdown.
- When storing the controller, seal it in a plastic bag with a desiccant inside the bag to protect from rust and dust.

3. Installation

The controller should be installed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with the robot arm	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with the robot arm	
Protective gloves	Protection part: Hand and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

⚠ DANGER

- Be sure to always install the controller outside of the robot's maximum space where you can see the robot's motion. If you perform operations within the robot's maximum space, the robot could collide with you.
- Mount a cover or other protection for the buttons and switches of the control unit connected to the controller. Accidental contact could turn on a signal, causing the robot to move, resulting in a collision.

⚠ WARNING

- Never install the controller in locations that are exposed to water. Water could seep into the controller, causing circuits to short-circuit, and result in a fire. It could also result in electric shock due to electric leakage or other causes.
- Never place flammable objects near the controller. If sparks occur due to a breakdown, nearby flammable objects could ignite, resulting in a fire.

⚠ CAUTION

- Do not place objects on top of the controller. (Stacking is also prohibited.)
  - A fire could occur due to lack of ventilation space and heat buildup.
  - Because the design assumes that no objects are placed on top of the controller, the controller could collapse due to being unable to support the object, and crush your hands or feet.
- Do not install on an unstable location. The controller could fall and crush your hands or feet.
- Install the controller inside the control panel or in a location where sufficient distance can be provided so that it cannot come into contact with the operator. If the controller is installed within the operator's range of motion, the operator could fall over with arms or legs caught in a cable, or could receive an electric shock due contact with the energized parts.
- Be sure to always perform the wiring for the robot after installation is completed.
  - If the controller is installed or carried while the wiring is connected, a cable could be broken, resulting in an electric shock.
  - Also, your legs could get caught in a cable, causing you to trip and fall.
- Provide a dedicated duct or similar protection to protect the cable lead-in of the control panel. If the cable is routed without using a dedicated duct, the cable could break and cause a fire. It could also result in electric shock due to electric leakage or other causes.
- Display the execution mode of the respective controller on the operation device. If the operator does not know the execution mode, a collision could occur with the robot due to an unexpected movement.

NOTICE

- Never supply a current exceeding the rated output. Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts.
- Do not place objects on top of the controller. (Stacking is also prohibited.)
  The controller was designed assuming that objects would not be placed on top of it, and so the top could be unable to withstand the weight and collapse, damaging the controller.
- If the LAN and OUTPUT connectors will not be used, be sure to attach the connector cap. If these connectors do not have any protection, electrostatic charges could cause a breakdown.
- When installing the controller in a closed space, such as in a control panel, use air vent holes, forced ventilation, and indirect cooling to prevent heat buildup. If the controller is installed in a closed space, heat will be trapped, resulting in a breakdown.

3.1 Installation Environment

Table 3.1 shows the environment specifications for the controller.

Table 3.1 Controller Environment Specifications

Item	Specifications
Temperature	During operation: 0 to 40°C In storage state: -10°C to 50°C
Humidity	20% to 80% (no condensation) Do not install the controller in a location exposed to liquids such as water.
Altitude	1000 m or less
Vibrations	Do not install in a location exposed to excessive vibrations or physical impacts or shocks.
Dust	Ensure that there is no electric conductive dust. Consult with Shibaura Machine beforehand if the equipment will be used in a dusty environment.
Gas	Do not install in a location exposed to corrosive or flammable gas.
IP (Ingress Protection) rating	IP20
Overvoltage category	IEC 60664-1 Class III
Protection against electric shock	IEC 61140 Class I
Pollution degree	IEC 60664-1 Pollution Level 3
Sunlight	Do not install in a location exposed to direct sunlight.
Power noise	Do not install in a location exposed to excessive noise in the surrounding area.
Magnetic field	Do not install in a location exposed to devices that generate strong magnetic fields in the surrounding area.
Other ambient environmental requirements	Do not install in a location exposed to iron powder, oil, salt, or organic solvents. Do not install in a location where the equipment will be splashed with water or other liquids.

3.2 Dimensions

Figure 3.2 shows the external views of the controller. The weight of the controller body is 11 kg.

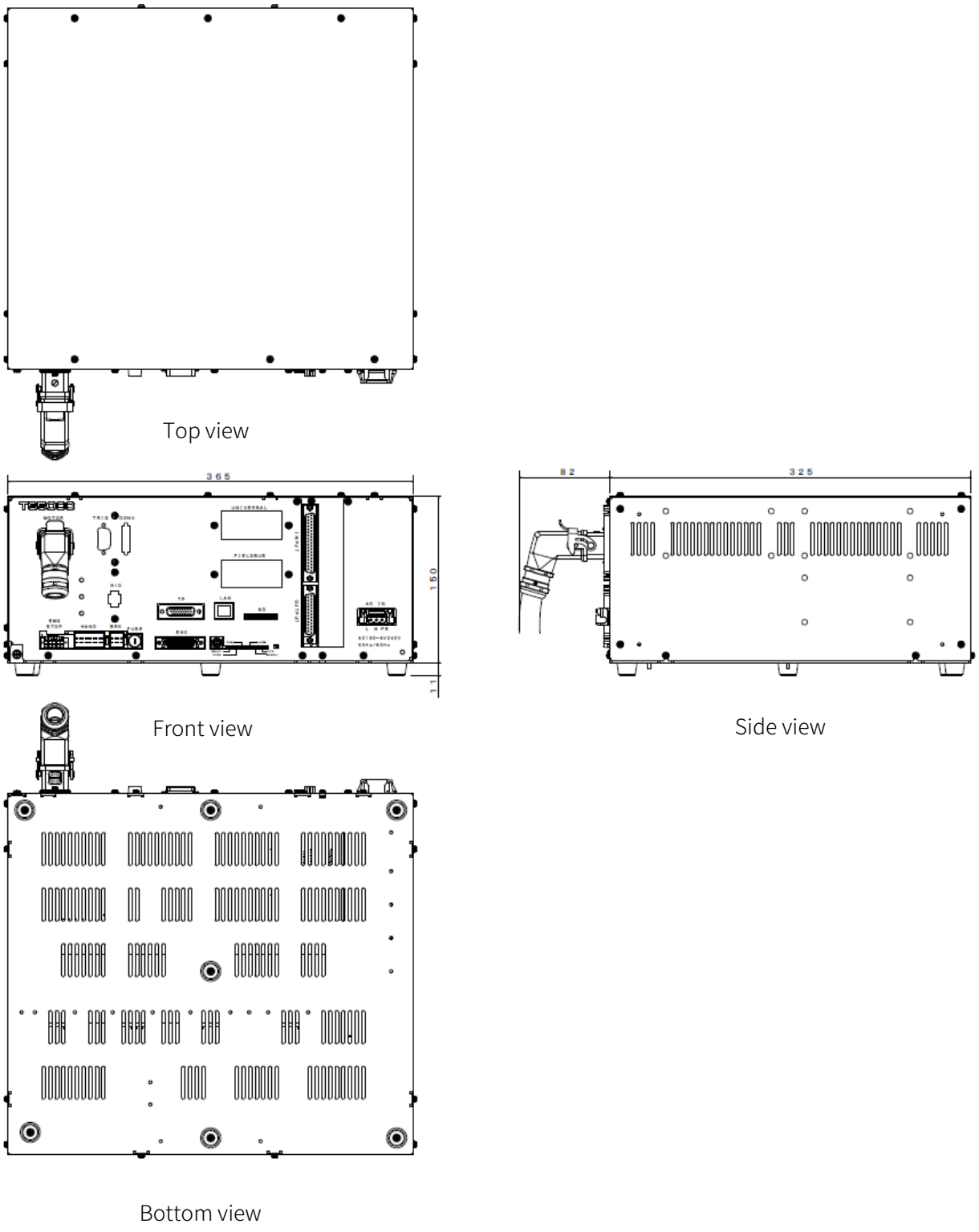


Figure 3.2 External Views

3.3 Safety Information for Installation on Floor

Figure 3.3 shows the required ventilation space for the controller.

⚠ WARNING

- When installing the controller in a closed space, such as in a control panel, use air vent holes, forced ventilation, and indirect cooling to prevent heat buildup. If the controller is installed in a closed space, heat will be trapped, and a fire could occur.

NOTICE

- Provide ventilation space around the controller of at least 50 mm on the right, left, and rear sides, 100 mm on the top side, and at least the height of the rubber feet (11 mm) on the bottom side. If no ventilation space is provided, the controller could overheat, resulting in a breakdown.
- If the controller will be installed in the control panel, use fixing jigs or other means to secure the controller in place. The controller could be exposed to vibrations, resulting in a breakdown.

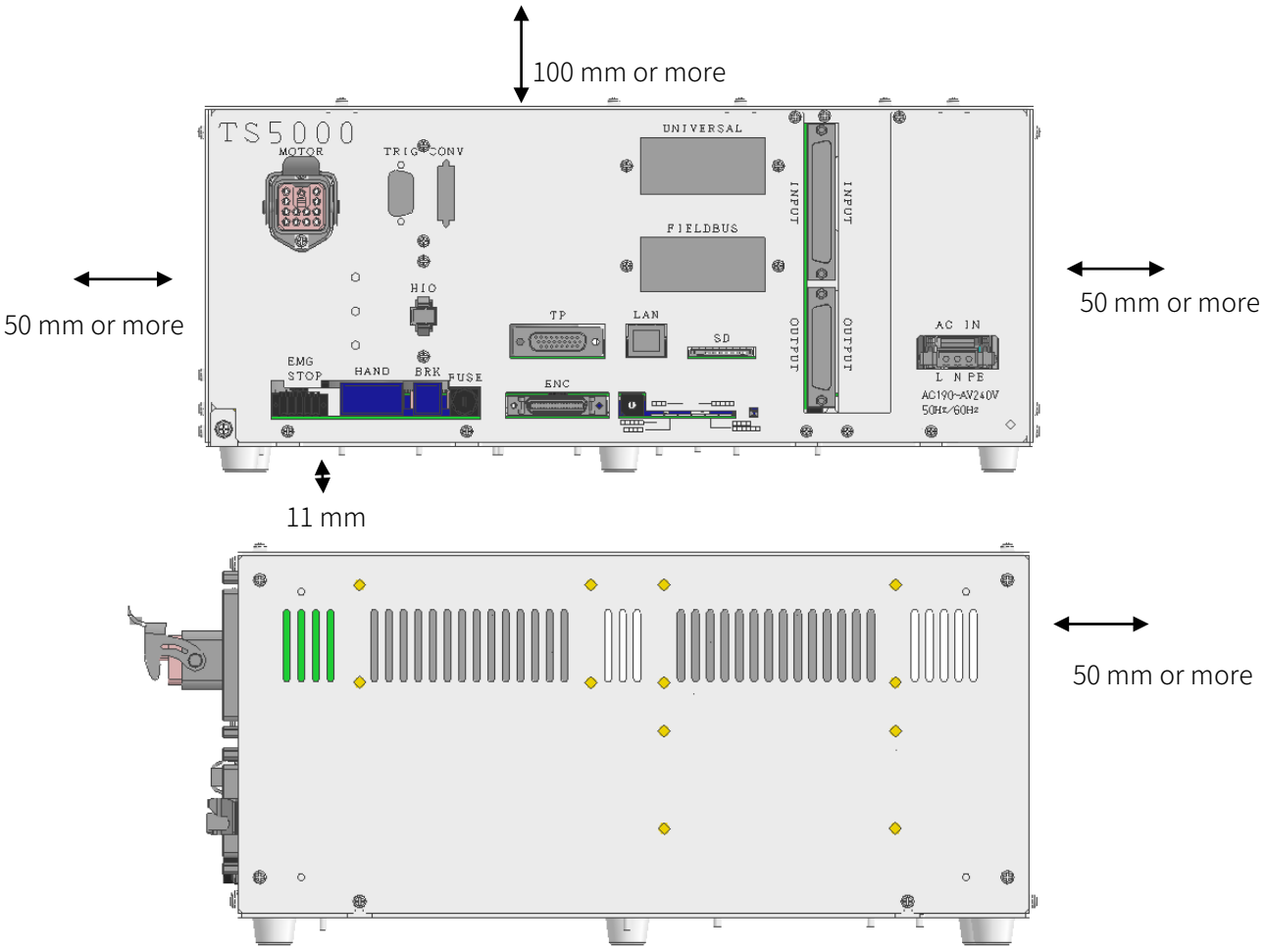
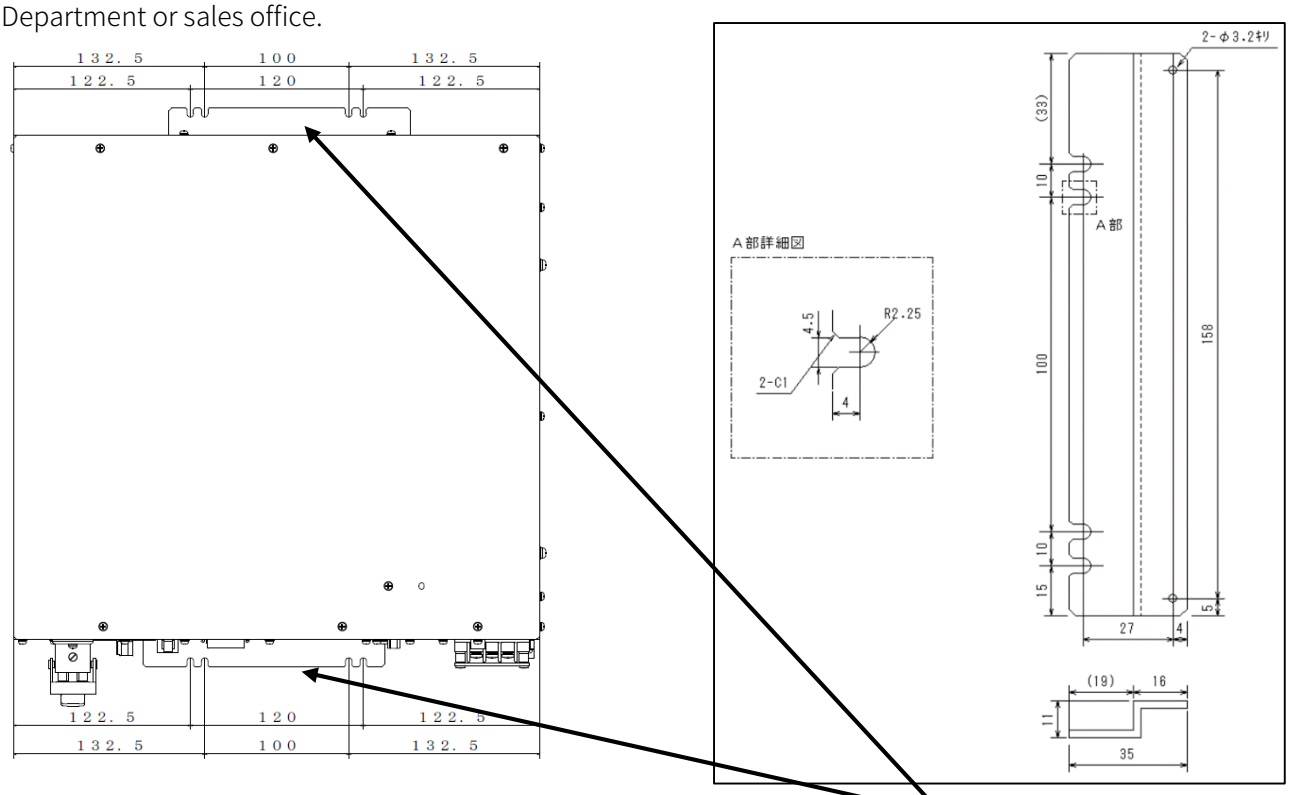


Figure 3.3 Required Ventilation Space for Controller

3.4 Mounting Surface Dimensions

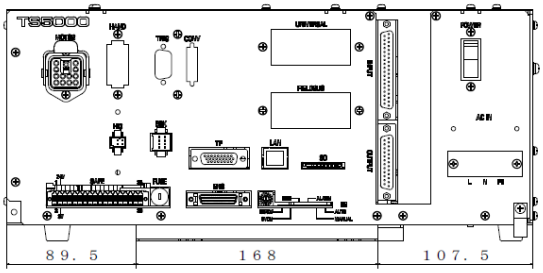
If the controller will be mounted to a control panel or other location, use a dedicated mounted fixture and secure the controller to the control panel with M4 screws. Also, the mounting work is the responsibility of the customer.

For more information about mounting brackets (option), please contact the Shibaura Machine Service Department or sales office.

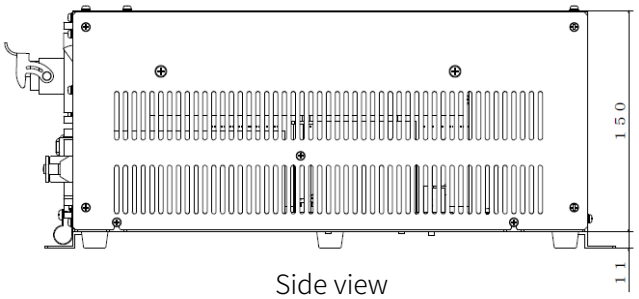


Top view

Mounting brackets



Front view



Side view

Figure 3.4 Controller Installation Dimensions

NOTICE

- If the controller will be installed in the control panel, use fixing jigs or other means to secure the controller in place. The controller could be exposed to vibrations, resulting in a breakdown.

3.5 Notes When Mounting to the Control Panel or Other Sections

Observe the following when mounting the controller to the control panel or other locations.

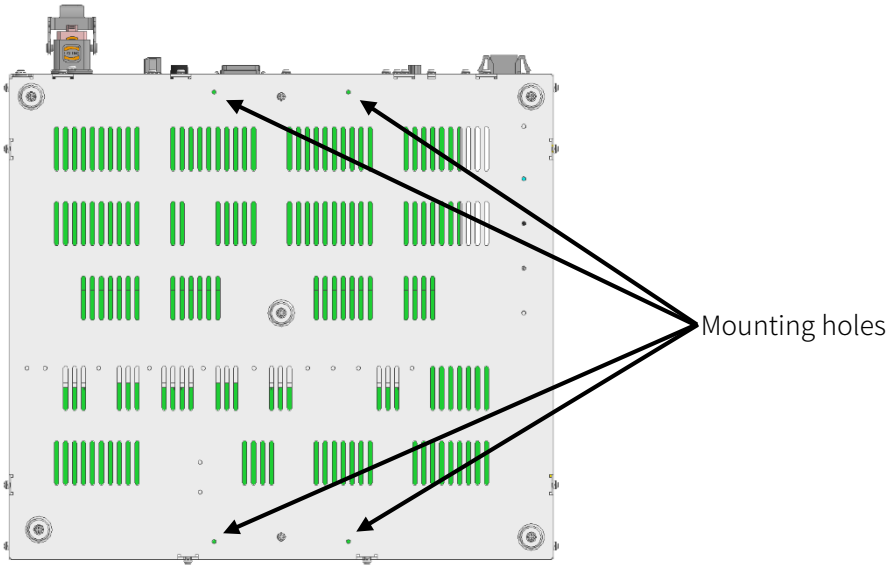
(a) When mounting the controller to the control panel or other location, remove the rubber feet from the bottom panel, and attach the mounting brackets to the mounting holes.

(b) When the controller is housed in the control panel, it must be taken out from the control panel when maintenance is performed. When mounting the controller, set the controller in a location where there are no impediments to performing maintenance of the controller.

(c) Arrange so that the cables connected to the controller can be pulled out.

(d) When taking out the controller from the control panel to perform maintenance, use a configuration where all cables can be reconnected, and the robot can be operated.

(e) Provide a clearance of 95 mm or more to enable the connection of the robot's cable connectors to the controller's front section.



Bottom panel view

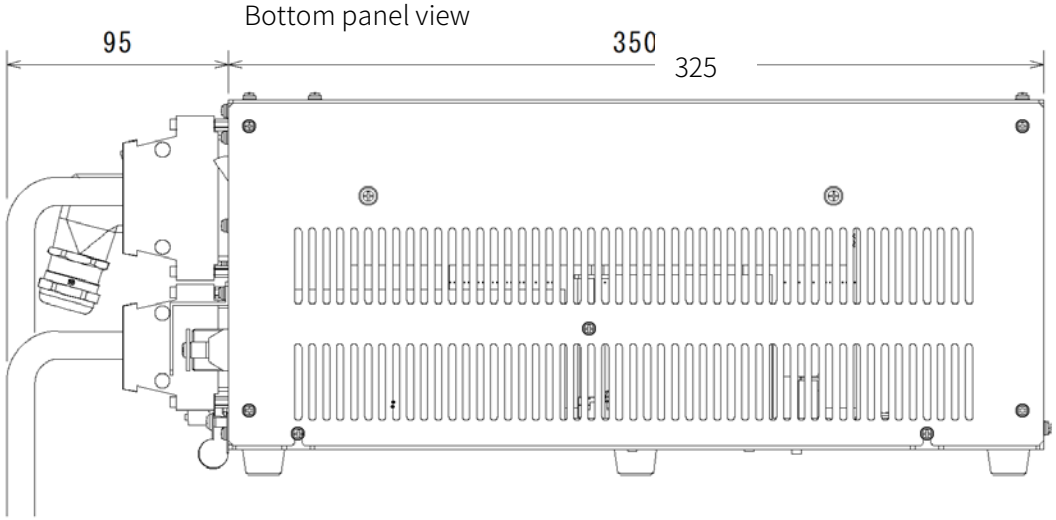


Figure 3.5 Clearance from Front of Controller

4. Cable Connection

Cables should only be connected by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with the robot arm	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with the robot arm	
Protective gloves	Protection part: Hand and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

⚠ DANGER

- Never use unauthorized cables. Fabricating and use of unauthorized cables can result in a fire.
- Do not connect to unauthorized connectors or terminals. Connecting to unauthorized connectors or terminals could cause a fire.
- Be sure to always connect a ground wire. Usage without connecting the ground wire can result in ground leakage, electric shock, or a fire.
- After wiring, use a tester or other device to check the voltage and for any short-circuits. If the power is turned on while the wiring is incorrect or while the controller is malfunctioning, a fire could occur. It could also result in electric shock due to electric leakage or other causes.
- Be sure to connect all connectors securely. If the connectors do not fit firmly, electrostatic charges could build up and lead to a malfunction, which could result in collision with a robot.

NOTICE

- Do not connect to unauthorized connectors or terminals. Unauthorized connectors and terminal wires could result in a breakdown of the controller.

4.1 Wiring

This section describes types of cables and connectors and how to connect them.

4.1.1 Connectors on the controller

Figure 4.1 shows the locations of the connectors on the controller. The following sections describe the connections for cables ① to ⑦. For details on the connector pin location and other cable connections, refer to the STEA1138 TS5000-SS Interface Manual.

① "ACIN" connector for connecting AC power cable
② "MOTOR" connector for connecting robot drive power cable
③ "HAND" connector for connecting robot hand I/O cable
④ "BRK" connector for connecting brake cable
⑤ "ENC" connector for connecting encoder cable
⑥ "EMGSTOP" connector for connecting input/output external power supply and external emergency stop input/output
⑦ TP cable

Robot control cables

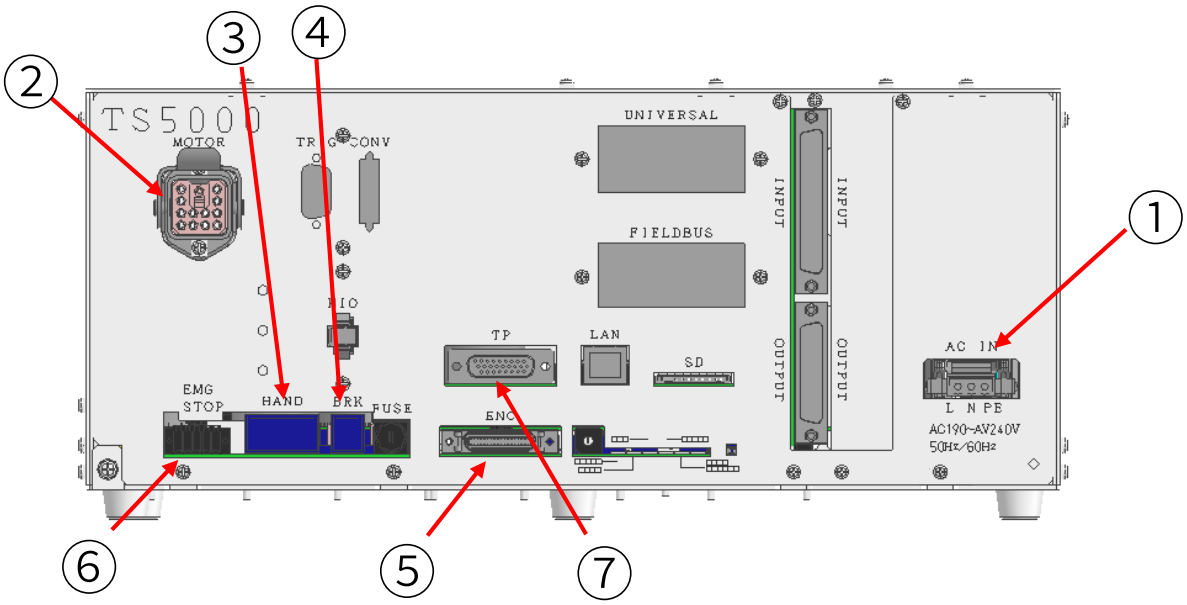


Figure 4.1 Controller Connector Locations

4.1.2 "ACIN" connector for connecting the power cable Figure 4.1-①

The power cable is used for supplying 200 V AC to the controller. The following plug is supplied. The cable is not included in the accessories. It must be manufactured by the customer. An ACIN connector is used (Figure 4.1-①).

Also, if the main power is not supplied properly to the controller due to a missing phase, low voltage, or other abnormality, the error "8-027 Slow Charge Error" occurs when the servo is turned on. If this alarm is issued, verify whether the supply voltage at the AC power connector satisfies the input power supply specifications value for the controller and whether the supply voltage is stable. For details, refer to STEA1165_Alarm Manual, 8-027 Slow Charge Error.

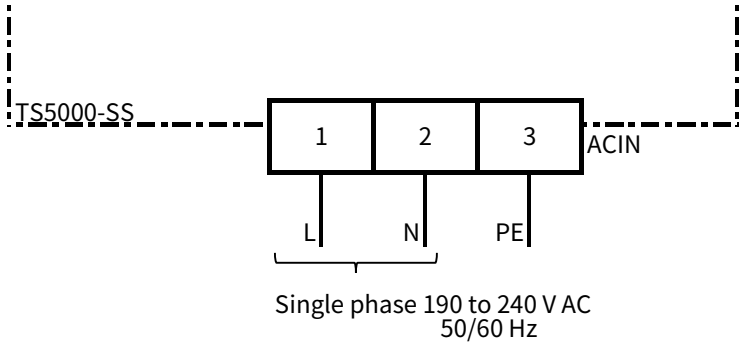


Figure 4.2 AC Power Cable Wiring

Table 4.3 Power Supply Specifications

Power supply	Single-phase, 190 to 240 V AC 50/60 Hz±1 Hz
Momentary power loss	Within 40 msec
Grounding	Class D only
Plug connector	Manufacturer: WAGO Kontakttechnik GmbH & Co. KG Model: 231-103/037-000
Recommended wiring	AWG18 to AWG14

⚠ DANGER

- Do not insert conductive objects or your fingers inside the power supply terminals. An electric shock could occur.

4.1.3 Robot control cable Figure 4.1-② to ⑤

The controller and robot are connecting using the robot control cables in " 4.1.3.1 "MOTOR" connector for connecting robot drive power cable Figure 4.1-②", "4.1.3.2 "HAND" connector for connecting robot hand I/O cable Figure 4.1-③", "4.1.3.3 "BRK" connector for connecting brake cable Figure 4.1-④", "4.1.3.4 "ENC" connector for connecting encoder cable Figure 4.1-⑤" and Figure -ⓐ.

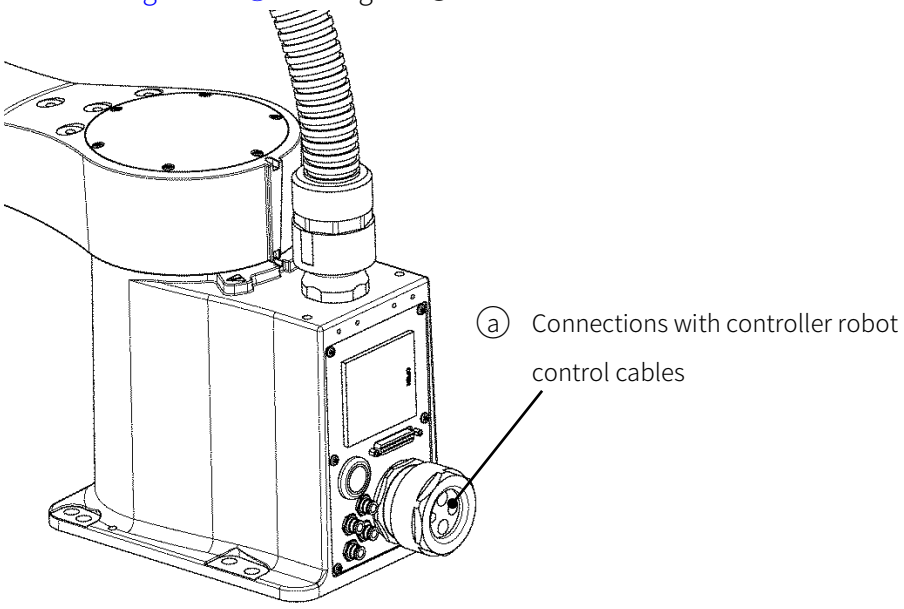


Figure 4.4 Robot-side Connector Locations

4.1.3.1 "MOTOR" connector for connecting robot drive power cable Figure 4.1-②

This cable connects the controller and robot, and it supplies the driving power needed for motor rotation to the various axis motors of the robot from the controller servo driver.

4.1.3.2 "HAND" connector for connecting robot hand I/O cable Figure 4.1-③

The hand I/O cable controls the hand on the robot tip, and in its standard configuration, is a signal cable capable of using 8 inputs and 8 outputs.

NOTICE

- Set the robot hand output signal to a maximum of 0.1 A per output (max. of 0.8 A for total 8 outputs). Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts.

4.1.3.3 "BRK" connector for connecting brake cable Figure 4.1-④

The brake cable turns on and off the brake for holding the motor axes.

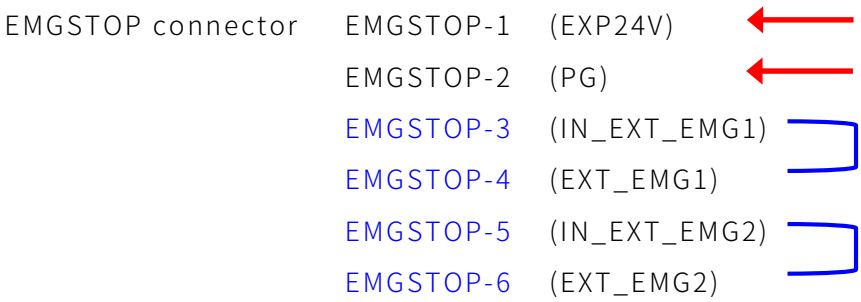
4.1.3.4 "ENC" connector for connecting encoder cable Figure 4.1-⑤

The encoder cable is a signal line to input signals from the rotation angle detection encoder of each robot axis to the controller.

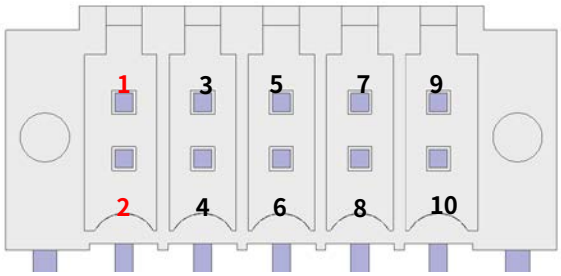
4.1.4 "EMGSTOP" connector for connecting input/output external power supply and external emergency stop input/output Figure 4.1-⑥

The EMGSTOP connector supplies power for I/O and emergency stop signal input to the controller from an external device. The supplied connector B2CF3.50/10/180LR is used for the emergency stop signal input or for the supply of 24 VDC from an external device. The I/O external power supply is used for system I/O, general-purpose I/O, extension I/O, and hand I/O. Because the connector does not have any printed pin numbers, see Figure 4.5 to connect the wires.

Also, if the emergency stop input/output from an external device will not be used, the emergency stop line of this connector must be connected with a jumper to enable robot operation. Wire by following the method shown in Figure 4.5. Four cables for the SAFF connector jumpers are included in the accessories. To install an external emergency stop switch, refer to STEA1238_TS5000-SSInterface Manual, 4.2 Connecting Input/Output External Power Supply Lines.



Controller side



Connector side

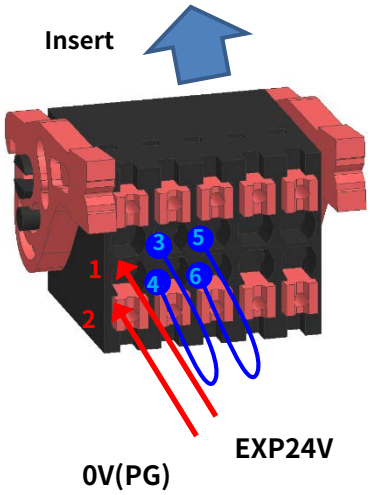


Figure 4.5 Jumper Connection Method for Standard Specifications

Table 4.6 I/O External Power Supply Specifications

Power supply	24 V DC±10%
Supported cables	0.14 mm^2 to 1.5 mm^2 (AWG26 to AWG16)

NOTICE

- Apply a voltage in a range of 24 V±10% to the input/output external wire. Applying a voltage exceeding the allowable value may damage the controller.

4.1.5 Connecting the TP connector Figure 4.1-⑦

The TP connector is a connector used to connect a teach pendant to the controller. The optional TP5000 and TP1000 (requires a dedicated conversion cable) can be connected.
For details on the TP connector, refer to STEA1238_TS5000-SS Interface Manual, 7. TP Cable.

MEMO

1. If the TP connector is disconnected while in AUTO mode, an emergency stop will occur and Master mode will be held in AUTO mode.

4.1.6 Connecting and disconnecting connectors

⚠ WARNING

- Turn off the power before connecting and disconnecting cables from the controller. Touching the terminals of any cables that were disconnected while power was flowing could result in an electric shock.
- When disconnecting the cable, pull it out by grasping the connector while holding down the controller.
  - Pulling the cable itself could break the cable and result in an electric shock.

NOTICE

- Turn off the power before connecting and disconnecting cables from the controller. Touching the terminal of a disconnected cable while the power is on could result in an electric shock, and connecting and disconnecting the cable while the power is on could damage the ICs in the board.
- When disconnecting the cable, pull it out by grasping the connector while holding down the controller.
  - The controller could fall over, resulting in a breakdown.

Otherwise, after the cable-side connector is connected to the controller, use the anchoring fixture to lock the connector connection in place. If the inserted connector is loose, this could lead to poor contact of the connector, resulting in an accident, and so check that it is firmly connected.
To remove the connector, remove the anchoring fixture, and then pull off the connector.

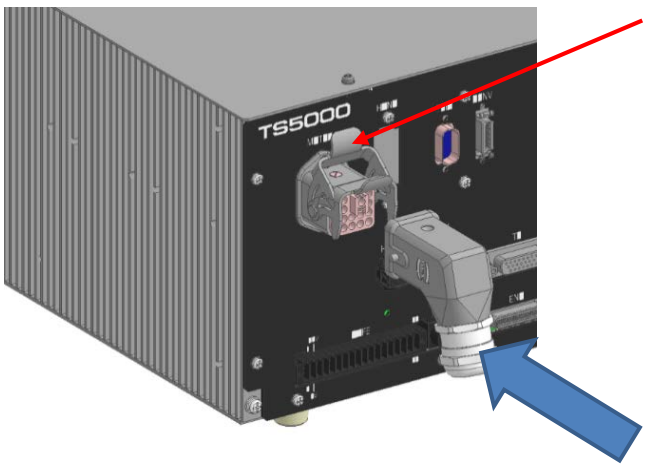


Figure 4.7 Connecting or Disconnecting the Robot Drive Power Cable

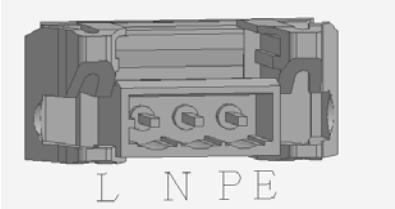
4.1.7 Connector terminal arrangement

4.1.7.1 AC power connector “ACIN” Figure 4.-①

Controller side

Model: 231-603/114-000

Manufacturer: WAGO



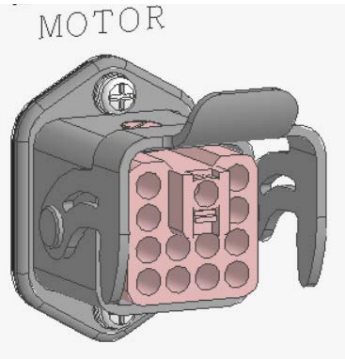
4.1.7.2 Robot drive power cable connector “MOTOR” Figure 4.-②

Controller side

Model: 09120123101

09200030301

Manufacturer: Harting Co., Ltd.

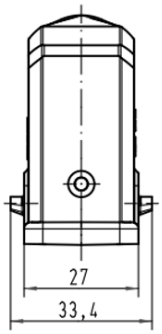


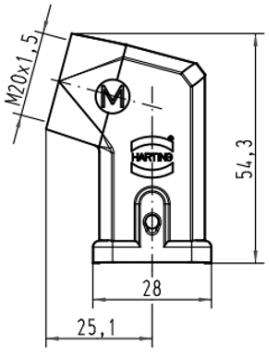
Cable side

Model: 19200031640

09120123001

Manufacturer: Harting Co., Ltd.



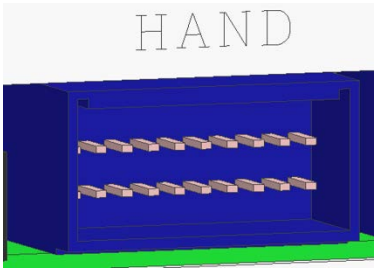


4.1.7.3 Robot hand I/O cable connector “HIO” Figure 4.-③

Controller side

Model: S20B-J21DK-GGXR

Manufacturer: J.S.T. Mfg. Co., Ltd.



Cable side

Model: J21DF-20V-KX-L

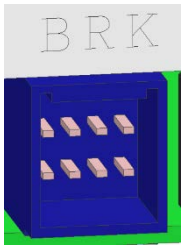
Manufacturer: J.S.T. Mfg. Co., Ltd.

4.1.7.4 Brake cable connector “BRK” Figure 4.-④

Controller side

Model: S08B-J21DK-GGXR

Manufacturer: J.S.T. Mfg. Co., Ltd.



Cable side

Model: J21DF-08V-KX-L

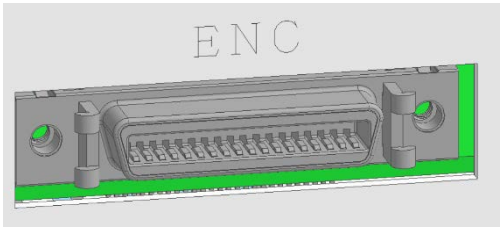
Manufacturer: J.S.T. Mfg. Co., Ltd.

4.1.7.5 Encoder cable connector “ENC” Figure 4.-⑤

Controller side

Model: 10236-59A3 MB

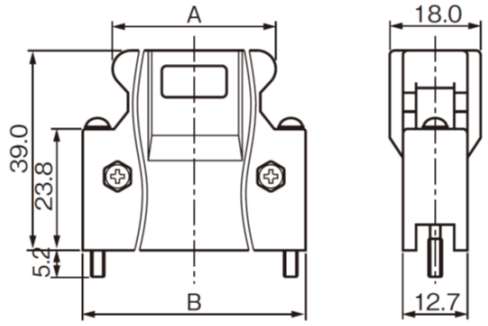
Manufacturer: 3M Company



Cable side

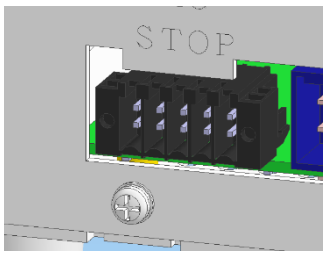
Model: 10336-52A0-008

Manufacturer: 3M Company



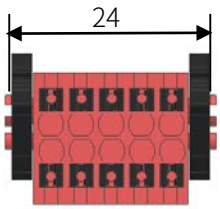
4.1.7.6 External emergency stop input/output and I/O external power cable connector “EMGSTOP” Figure 4.-⑥

Controller side
Model: S2C-SMT3.50/10/90LF
Manufacturer: Weidmüller Interface GmbH & Co. KG



The mating connector is supplied to the controller. For details on the cable, see 4.1.4 "EMGSTOP" connector for connecting input/output external power supply and external emergency stop input/output Figure 4.1-⑥, or refer to STEA1238_TS5000-SS Interface Manual, 4.1 Input/Output External Power Supply Cable.

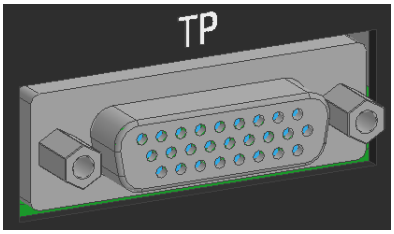
Cable side
Model: B2CF3.50/10/180LR
Manufacturer: Weidmüller Interface GmbH & Co. KG



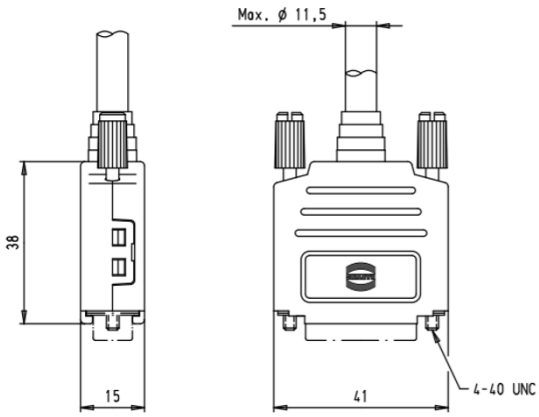


4.1.7.7 TP Connector “TP” Figure 4.-⑦

Controller side
Model: 09 56 252 7613
Manufacturer: Harting Co., Ltd.



Cable side
Model: 09562005604
Manufacturer: Harting Co., Ltd.



4.2 Controller INPUT

4.2.1 Connector signal connection diagram

For a detailed drawing of the connector terminals, refer to STEA1238_TS5000-SS Interface Manual, 2. Connecting External Connection Cables.

4.2.2 INPUT jumper connector

The following system input signal lines are provided as safety measures. The robot will not run unless a jumper connection is made for the following signal lines. Consequently, INPUT jumper connectors[1)] where jumper connections have been made are included for the following system input signals. The internal jumper when this connector is used for the TS5000-SS is a P24G(-) common type P.

1) Product name: I/O jumper connector.

System input signal line INPUT-12 (STOP)
INPUT-14 (SVOFF)
INPUT-32 (BREAK)
INPUT-16 (P24V)
INPUT-17 (SYSINCOM)
INPUT-35 (P24G)

Jumper connections for attached connectors

INPUT (Pin No.)					
12-16	14-16	32-16	17-35	18-19[1)]	36-37[1)]

1) This jumper connection is not used in this controller.
The robot can be run without making this jumper connection.

Also, use jumper connections for the following signals if they will not be used as system signals.

INPUT-13 (LOW_SPD)
INPUT-31 (CYCLE)
INPUT-16 (P24V)

MEMO

1. *If a jumper connection is not used for the "STOP" signal, robot automatic operation cannot be performed.*
2. *If a jumper connection is not used for the "SVOFF" signal and emergency stop contact 1 and 2 signals, the servo power supply of the controller cannot be tuned on.*
3. *If a jumper connection is not used for the "BREAK" signal , robot automatic operation cannot be performed.*
4. *If a jumper connection is not used for the "LOW_SPD" signal, the robot automatic operation speed will be slow speed.*
5. *If a jumper connection is not used for the ”CYCLE” signal, the controller will switch to cycle operation mode.*
6. *For the connectors supplied to the controller, jumper connections are already used for the signals in 1 to 3. However, note that the connection is P24G(-) common type P. Also, to operate the robot without using the system input signal, connect the supplied INPUT jumper connector to the INPUT connector on the controller.*
7. *If an input signal cable will be fabricated by the customer, either change the INPUT jumper connector wiring or obtain the cable in STEA1238_ TS5000-SS Interface Manual, 13. INPUT Cable.*

5. Operation Check

The controller must be operated by a qualified technician. These processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with the robot arm	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with the robot arm	
Protective gloves	Protection part: Hand and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

5.1 Controller operation check

・Startup check

Check that the controller is starting normally.

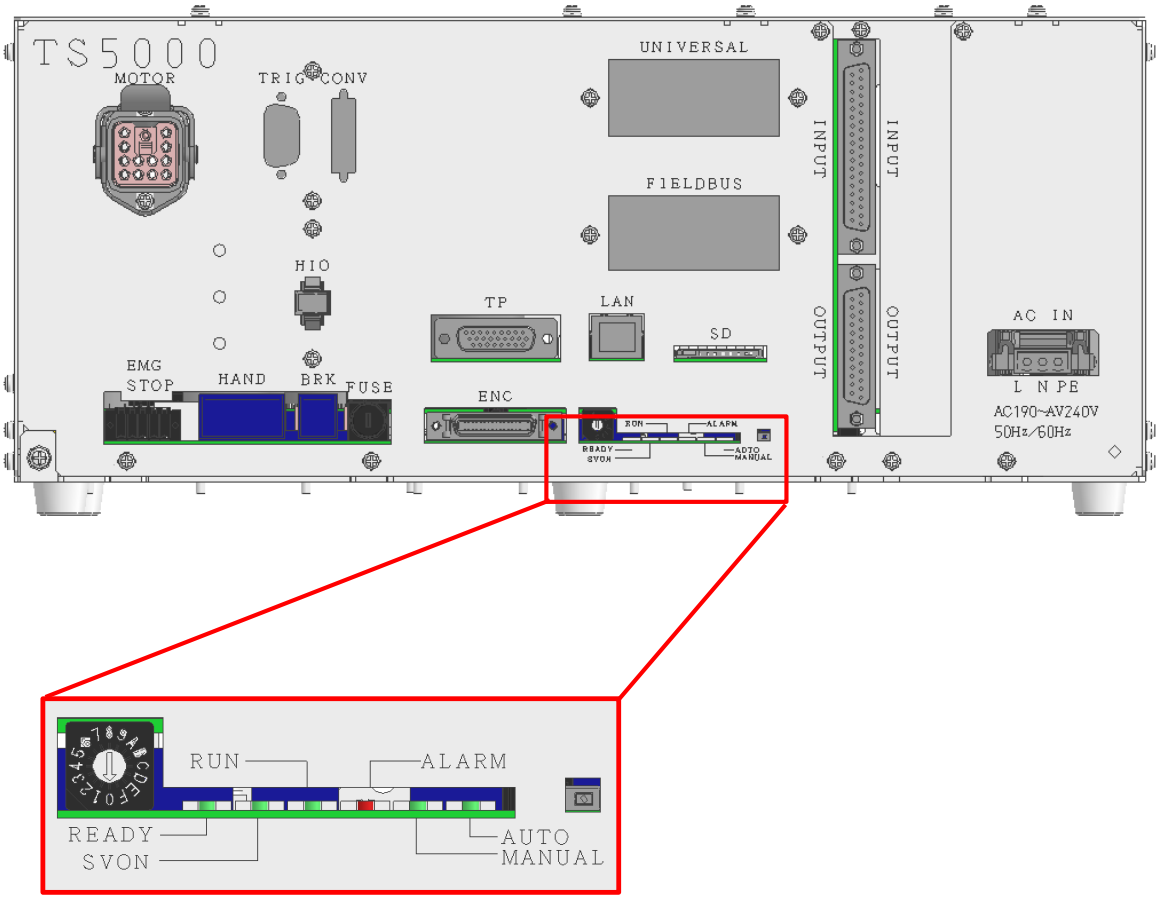
Before turning on the power of the controller, make sure that the ACIN is connected to the single-phase 200 V and the 24 VDC is connected to the EMGSTOP connector, and then turn on the power supply circuit breaker. If the controller started normally, the “READY” LED on the bottom front of the controller is turned on.

If the teach pendant is connected and the controller is started up normally, the main screen will be displayed after startup.

・Reboot

Check that the controller can be rebooted normally.

Once you are ready to turn off the controller power, turn off the power supply circuit breaker, and then turn off the power. Then, turn on the power supply circuit breaker again. If the controller started normally, the “READY” LED on the bottom front of the controller is turned on.



LED display unit

MEMO

For details, refer to STEA1162 TP5000 Operation Manual.

5.2 Safety Information for Teaching Operations

The teaching operations should be performed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country.

⚠ DANGER

- Be sure to always perform the operations as described in the instruction manual. Incorrect operation could result in a collision with the robot. An electric shock could also occur.
- Be sure to set up a safety fence or door, and perform teaching outside the robot's maximum space. If you perform teaching within the robot's maximum space, the robot could collide with you.
- When turning on the servo, be sure to always be outside the robot's maximum space. Otherwise, a collision could occur due to an unexpected movement by the robot.

⚠ WARNING

- Never enter the robot's maximum space during test operation.
  A collision could occur with the robot.
- Never leave any equipment in the work space that could interfere with your work.
  This could cause you to trip and fall. If you concentrate on teaching, you could fail to pay attention to your surroundings and collide with peripheral devices.
- Never open the cover of the controller during operation. Do not connect or disconnect connectors. An electric shock could occur.
- Never install the controller in locations that are exposed to water. Water could seep into the controller, causing circuits to short-circuit, and result in a fire. It could also result in electric shock due to electric leakage or other causes.
- Never place flammable objects near the controller. If sparks occur due to a breakdown, nearby flammable objects could ignite, resulting in a fire.
- Do not touch the controller while the power is turned on and for a while after the power is turned off. A burn could occur.

⚠ CAUTION

- All hand output signals are in an OFF state when the controller power is shut off and turned on. Design the end effector so that objects will not drop from the hand due to a power outage or similar situation. Objects dropped from the end effector could crush the hands or feet of the operator.

NOTICE

- Do not place any obstacles within the robot's work envelope. A collision could occur with the robot.
- Set the output signal to a maximum of 0.1 A per output. Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts.
- Supply a voltage in a range of 24 V±10% to the external input/output power supply (24 V DC). Applying a voltage exceeding the allowable value may damage the controller and result in a fire.
- Be sure to turn off the servo power while the robot is stopped before turning off the main power. Otherwise, if the main power is turned off while the robot is operating, the arm will run empty and collide with peripheral devices.
- Always check the execution status of the program when performing automatic operation. Failure to perform a check could result in interference with the peripheral devices and a breakdown of the equipment.

- If there are multiple controllers in the equipment, ensure that they operate the robot using the same operation mode. If the operation modes are different, a collision could occur between robots due to an unexpected movement.
- Be sure to turn off the power if an abnormality occurs when the equipment is turned on or if the controller's "READY" LED does not light up. The controller may be malfunctioning, and so please contact the Shibaura Machine Service Department.
- Stop operation if there is any unusual noise or odor from the controller or robot, or if it behaves differently than expected. The controller may be malfunctioning, and so please contact the Shibaura Machine Service Department.
- Be sure to always perform daily inspection before operating the equipment. Failure to conduct daily inspection could result in a breakdown of the controller.
- Perform periodic maintenance and inspection. Failure to conduct periodic inspection could result in a breakdown of the controller.

5.3 Safety Information for Operation

The operations should be performed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country.

MEMO

Do not change from MANUAL mode to AUTO mode while the safety protection functions are deactivated. When changing from MANUAL mode to AUTO mode, make sure that the safety protection functions are restored.

⚠ DANGER

- Be sure to always perform operations as described in the instruction manual. Incorrect operation could result in a collision with the robot. An electric shock could also occur.
- Be sure to set up a safety fence or door, and perform teaching outside the robot's maximum space. If you perform teaching within the robot's maximum space, the robot could collide with you.
- When turning on the servo, be sure to always be outside the robot's maximum space. Otherwise, a collision could occur due to an unexpected movement by the robot.

⚠ WARNING

- Never enter the robot's maximum space during test operation.
  A collision could occur with the robot.
- Never leave any equipment in the work space that could interfere with your work.
  This could cause you to trip and fall. If you concentrate on teaching, you could fail to pay attention to your surroundings and collide with peripheral devices.
- Never open the cover of the controller during operation. Do not connect or disconnect connectors. An electric shock could occur.
- Never install the controller in locations that are exposed to water. Water could seep into the controller, causing circuits to short-circuit, and result in a fire. It could also result in electric shock due to electric leakage or other causes.
- Never place flammable objects near the controller. If sparks occur due to a breakdown, nearby flammable objects could ignite, resulting in a fire.
- Do not touch the controller while the power is turned on and for a while after the power is turned off. A burn could occur.

⚠ CAUTION

- All hand output signals are in an OFF state when the controller power is shut off and turned on. Design the end effector so that objects will not drop from the hand due to a power outage or similar situation. Objects dropped from the end effector could crush the hands or feet of the operator.

NOTICE

- Do not place any obstacles within the robot's work envelope. A collision could occur with the robot.
- Set the output signal to a maximum of 0.1 A per output. Supplying a current that exceeds the rated output will damage the output elements, circuit board, and other parts.
- Supply a voltage in a range of 24 V±10% to the external input/output power supply (24 V DC). Applying a voltage exceeding the allowable value may damage the controller and result in a fire.
- Be sure to turn off the servo power while the robot is stopped before turning off the main power. Otherwise, if the main power is turned off while the robot is operating, the arm will run empty and collide with peripheral devices.
- Always check the execution status of the program when performing automatic operation.
  Failure to perform a check could result in interference with the peripheral devices and a breakdown of the equipment.
- If there are multiple controllers in the equipment, ensure that they operate the robot using the same operation mode. If the operation modes are different, a collision could occur between robots due to an unexpected movement.
- Be sure to turn off the power if an abnormality occurs when the equipment is turned on or if the controller's "READY" LED does not light up. The controller may be malfunctioning, and so please contact the Shibaura Machine Service Department.
- Stop operation if there is any unusual noise or odor from the controller or robot, or if it behaves differently than expected. The controller may be malfunctioning, and so please contact the Shibaura Machine Service Department.
- Be sure to always perform daily inspection before operating the equipment. Failure to conduct daily inspection could result in a breakdown of the controller.
- Perform periodic maintenance and inspection. Failure to conduct periodic inspection could result in a breakdown of the controller.

Maintenance

This chapter describes the maintenance and inspection of the controller.

6. Maintenance

Controller maintenance and inspection must be performed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with the robot arm	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with the robot arm	
Protective gloves	Protection part: Hand and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

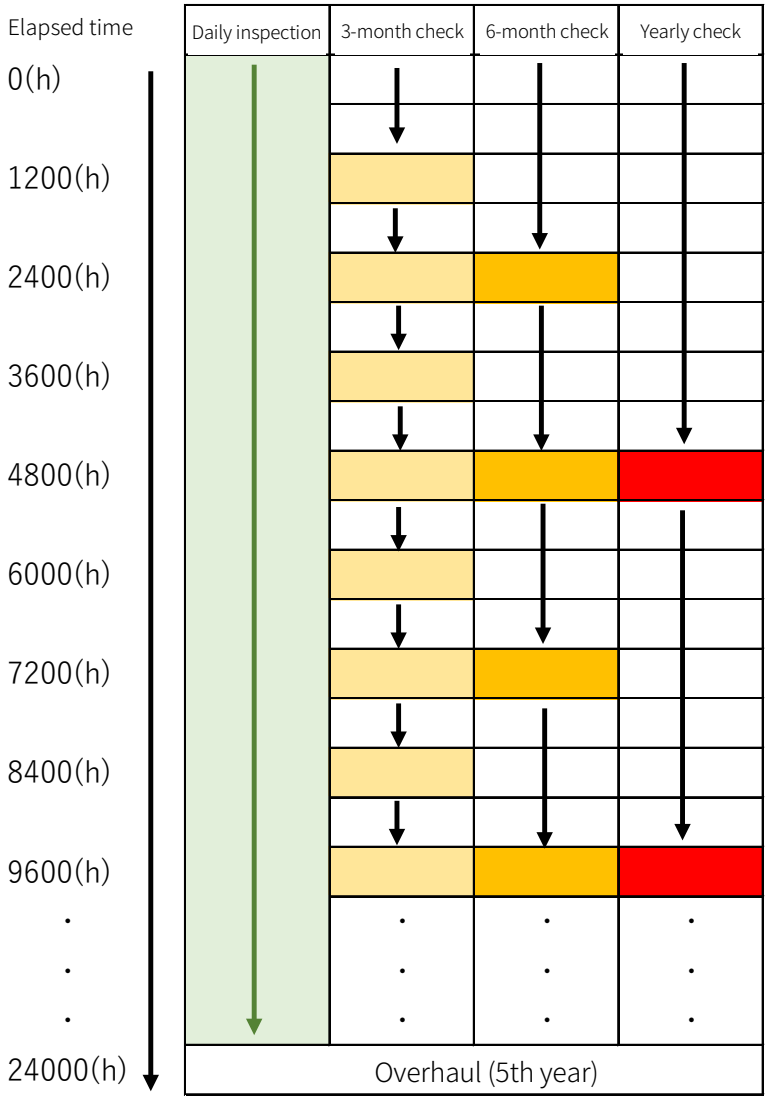
NOTICE

- Perform periodic maintenance and inspection. Failure to conduct periodic inspection could result in a breakdown of the controller.

6.1 Maintenance Schedule

There are two types of maintenance and inspections: Those conducted daily and those performed after a certain period of time.

In the periodic inspection and maintenance, inspection items are added in 1,200-hour units (hours of operation).



Periodic maintenance and inspection time calculation

16 operating hours per day: 1200 hours / 16 hours / 25 days = 3 months
24 operating hours per day: 1200 hours / 24 hours / 25 days = 2 months

When the system is operated for 16 hours per day, periodic inspection should be conducted every three months. However, if the hours of operation per day are increased, periodic inspection must

Figure 6.1 Maintenance Schedule

6.2 Maintenance and Inspection Items

There are two types of maintenance and inspections: Those conducted daily and those performed after a certain period of time. Inspection and replacement are described in their respective sections. For details about maintenance of the robot, refer to the STEA1234_THL500/THL600/THL700 Robot Manual or STEA1213_THE400/TS5000 Series Robot Manual.

6.2.1 Appearance inspection

Table 6.2 Inspection Items

Details of Inspection	Inspection Location	Daily Inspection	3-month Inspection	6-month Inspection	1-year Inspection	Refer to
Connector tightness	Entire connector section	○	○	○	○	4.1.1
Inspect for cuts or scratches Clean off any adhering foreign objects	The entire controller	→	○	○	○	
Vent hole check	Vent holes	→	→	○	○	7.5.1

6.2.2 Inspection during robot operation

Table 6.3 Inspection Items

Details of Inspection	Inspection Location	Daily Inspection	3-month Inspection	6-month Inspection	1-year Inspection	Refer to
Emergency stop operation check 2)	Emergency stop switch	○	*	*	*	7.5.2 7.5.3
Enable switch operation check	Enable switch	→	○	○	○	
Master mode switch MANUAL/AUTO mode operation check	MANUAL/AUTO mode	→	○	○	○	
Robot/controller operation check	Operation check of all axes in MANUAL mode	→	○	○	○	1)

1) Refer to STEA1162_TP5000 Operation Manual.

2) In the daily inspection, it is acceptable to perform the operation check while the servo is turned off, but an operation check while the servo is on should also be performed at least once a week.

* This is a daily inspection item, but for safety reasons, this is rechecked in the periodic inspection. If any abnormality is found in inspection, please contact the Shibaura Machine Service Department.

6.2.3 Overhaul

Table 6.4 Overhaul

Maintenance Description	Maintenance Parts	5-year	Refer to
Battery replacement	Controller battery	○	1)
Switching power supply replacement	Inside the controller	○	

1) For details about overhauls, please contact the Shibaura Machine Service Department.

6.3 Maintenance Contract and Repair Service

6.3.1 Maintenance contract

Maintenance such as overhauls is difficult for customers to perform, and so customers should request this work from the Shibaura Machine Service Department. We recommend that the customer enter into a service contract with Shibaura Machine at the time of purchase.

6.3.2 Repairs

If a fault has occurred or if repair is necessary, turn off the controller power and contact the Shibaura Machine Service Department. Provide a description of the fault and the following information indicated on the robot and controller.

(For contact information for the Shibaura Machine Service Department, see Service Network.)

- Controller model (MODEL)
- Controller's serial number (CONTROLLER S/N)
- Robot's serial number (ROBOT S/N)
- Controller's production date

For the serial number and manufacturing date, refer to the nameplate below on the controller top side. Also, for the controller model, see 1.5 Controller Model Name.

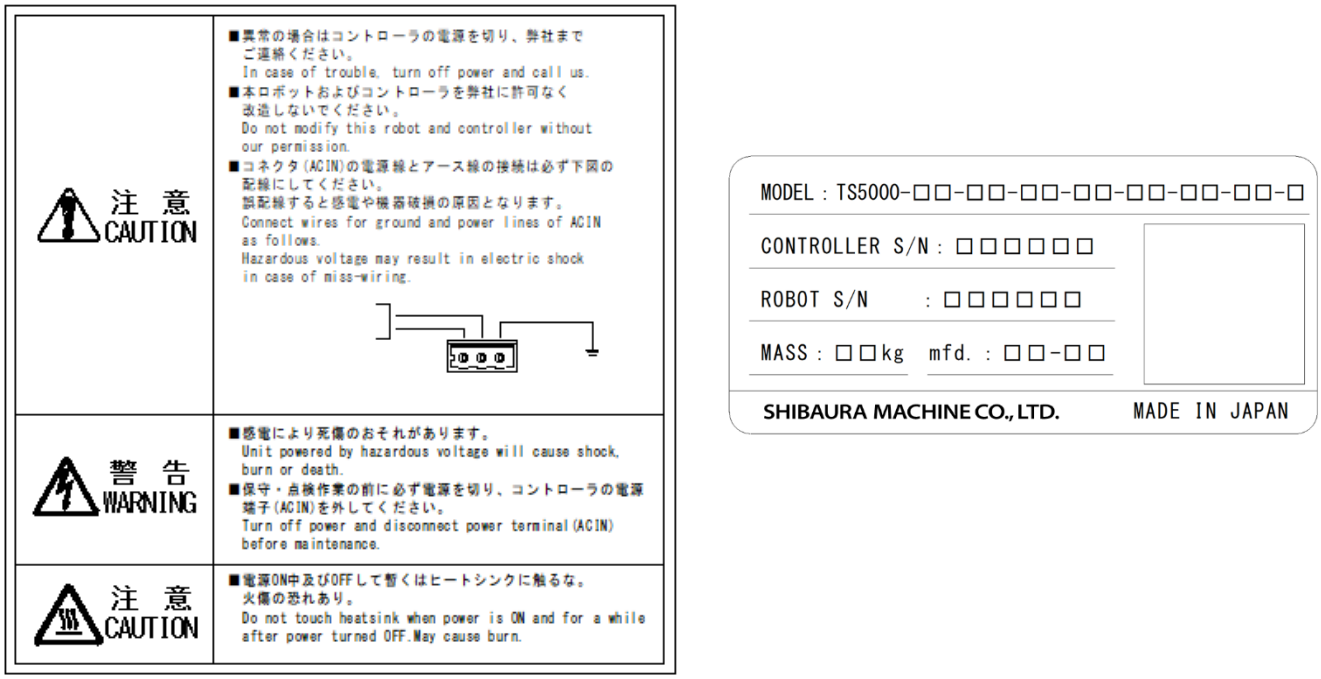


Figure 6.5 Nameplates

6.4 Modifications

Do not disassemble or modify the controller without the consent of Shibaura Machine

⚠ DANGER

- Never perform any parts replacement or modification except those described in the instruction manual. Reduced performance or malfunctions could result in electric shock, fire, or collision due to unexpected robot movements.

6.5 Maintenance Tools

We recommend that you provide the following tools for maintenance.

- #1 and #2 Phillips screwdrivers, #2 flat head screwdriver
- Nippers (to cut cable ties)

Recommended items besides tools:

- Alcohol (for cleaning)
- Cleaning cloth
- TSAssist program creation/teaching support software (option)

7. Maintenance Overview

Controller maintenance and inspection must be performed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country. The Shibaura Machine Service Department will perform replacement work. A fee is charged for this service. Be aware that Shibaura Machine's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

Recommended protector:

Type and name	Protection part and use	Recommended example
Helmet	Protection part: Head Use: Protect from a falling object Protect from collision with the robot arm	
Safety glasses	Protection part: Eyes Use: Protect from a flying object Protect from collision with the robot arm	
Protective gloves	Protection part: Hand and fingers Use: Protect hands and fingers when caught in the machine Prevent a carried object from dropping	
Safety shoes	Protection part: Feet and toes Use: Protect from a falling object	

7.1 Safety Information on Maintenance and Inspection

For maintenance and inspection of the controller, observe the following to ensure safety.

⚠ DANGER

- When opening the cover of the controller, do not allow water or other fluids or foreign objects to get inside. This could cause a short-circuit inside the controller, resulting in a fire. An electric shock could also occur due to electric leakage or other causes.
- Use SHIBAURA MACHINE special products for certain replacement parts for maintenance. Wiring of unauthorized connectors and terminals could cause a fire. An electric shock could also occur due to electric leakage or other causes.
- Be sure to always request part replacement from the Shibaura Machine servicing company for parts replacement that is not described in this manual. (A fee will be charged for this work.) Shibaura Machine will not be liable for any accidents or damage that arises due to replacement work performed by the customer or its results.

⚠ WARNING

- Wait about two minutes after disconnecting the power supply before touching the terminals of the controller or the internal circuit board. Residual voltage may remain in the high-capacity capacitors in the controller, which could result in an electric shock.
- Use Shibaura Machine-authorized spare parts when replacing parts. Use of unauthorized parts could cause a fire. A collision could occur with the robot due to an unexpected movement.
- Before starting any work, be sure to always wear protective gear, and unplug the power cable. If the power is turned on, the Power supply amplifier module, switching power supply, and other components will carry a charge, and an electric shock could result.
- Turn off the power before opening the controller cover. When opening the cover while the power is on, your hands or conductive tools could touch the current-carrying parts, resulting in an electric shock.
- Perform the following actions whenever entering the movable range of the robot.
  - Wear protective gear.
  - Place the teach pendant in an accessible location to the worker.
  - Change to MANUAL mode (remove the mode selector key so that the mode cannot be changed).

  ⇒If the above operations are not performed, the emergency stop could be cleared by another person during the daily inspection work within the movable range, and a collision could occur with a robot that was set to continuous operation.
- Use voltage measurement and the LEDs to check that no PN voltage remains in the large-capacity capacitor. If a PN voltage remains, an electric shock could occur.
- Do not allow conductive objects or hands to get in while the power is on. Touching the power-carrying part could result in an electric shock.
- All replacement parts for maintenance are parts with special specifications. Please request these parts from SHIBAURA MACHINE when purchasing and ordering.
- Wiring of unauthorized connectors and terminals could result in a fire. An electric shock could also occur due to electric leakage or other causes.

7.2 TP5000 Operation Restrictions

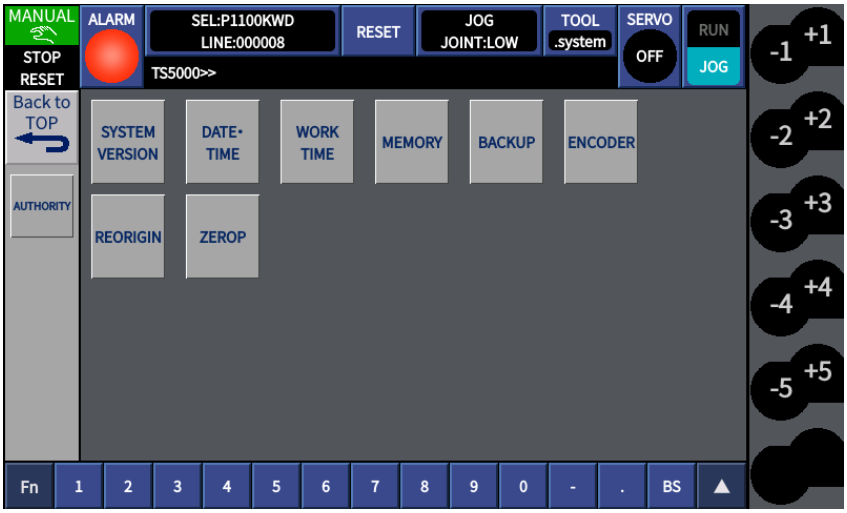
The TP5000 teach pendant has a screen operation restriction function to prevent data loss due to incorrect operation by a third party.

To perform maintenance operations on the maintenance screen, the user type must be switched to “MAINTE”.

Switching the user type to “MAINTE”

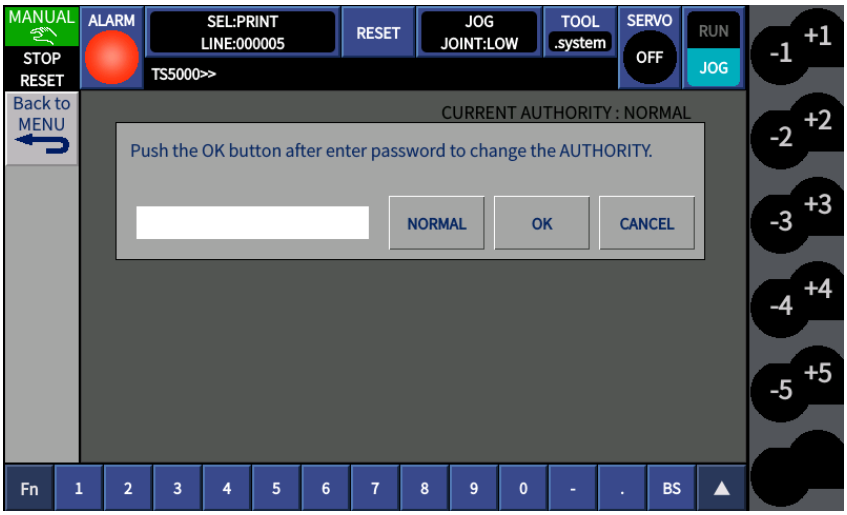
1 Tap the “MAINTE” button on main screen.

The maintenance menu screen appears.



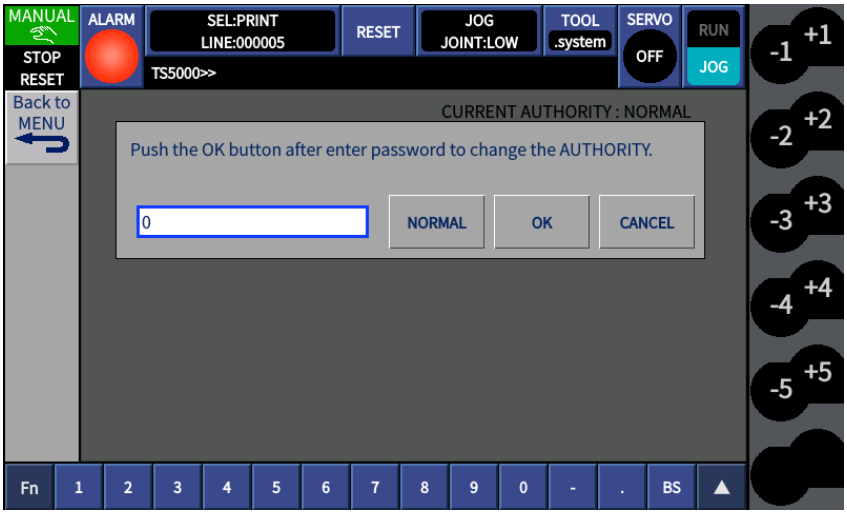
2 Tap the “Switch Users” button.

A password input window appears.



3 Tap the text box.

The cursor is displayed in the text box, indicating that it can be edited.



4 Use the keyboard to enter “0,” and tap the OK button.

The user type is changed to “MAINTE”.

> *MEMO*
>
> *The changed user type is valid until it is switched to another user type or until the TP5000 is turned off.*
>
> *When the TP5000 is powered on, the user type is set to “NORMAL”.*

7.3 Data Backup

Be sure to perform the controller maintenance after backing up the data.

7.3.1 Data backup (TP5000)

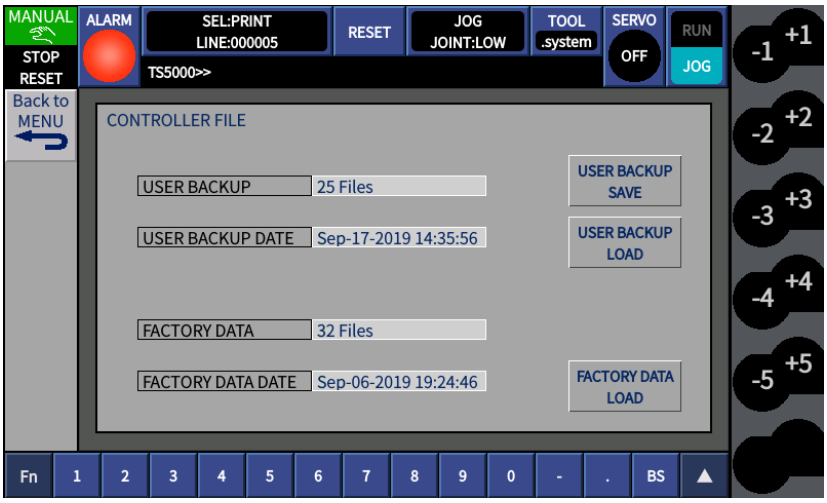
1 Once a TS5KBKUP folder is created in an SD card, after the controller is started up, insert the SD card into the TS5000 SD card slot.

2 Switch the user type to “MAINTE”.

For the detailed operating procedure, see "7.2 TP5000 Operation Restrictions".

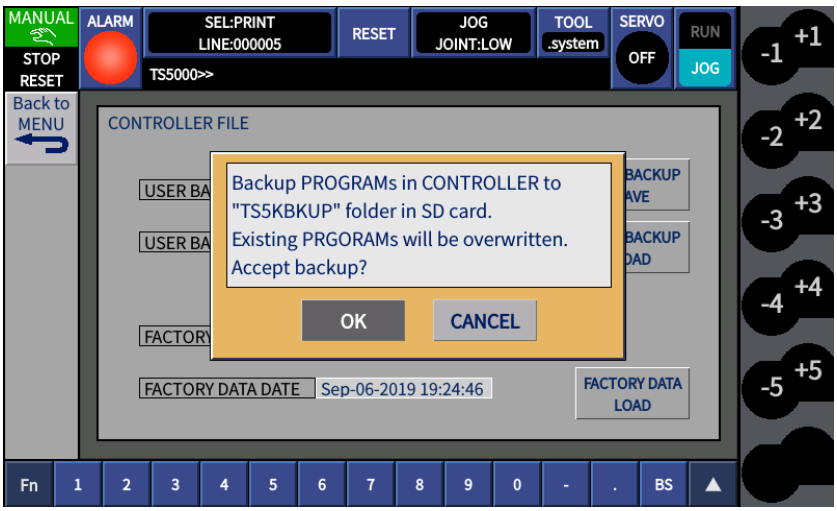
3 Press the “BACKUP” button on the maintenance screen.

The backup screen is displayed.



4 Tap the “Save Backup" button .

A backup confirmation screen is displayed.



5 Tap the OK button.

The current controller program files and parameter files are saved in the TS5KBKUP folder on the SD card.

NOTICE

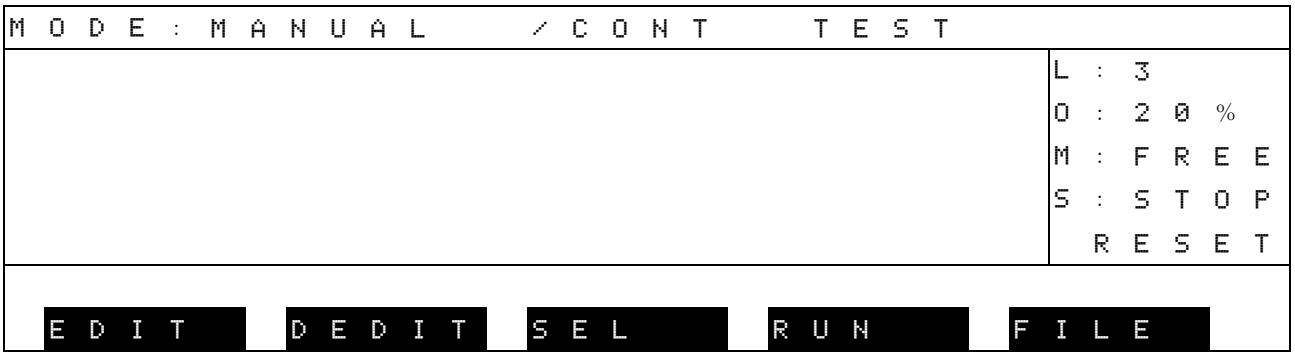
- The files in the TS5KBKUP folder of the SD card will be overwritten. To prevent overwriting of valuable data, move the target file from the TS5KBKUP folder to another storage location.

7.3.2 Data backup (TP1000)

This section explains how to back up files (parameter files and program files) using the TP1000. For other operating procedures using the TP1000, refer to the STEA1169_TP1000 Operation Manual.

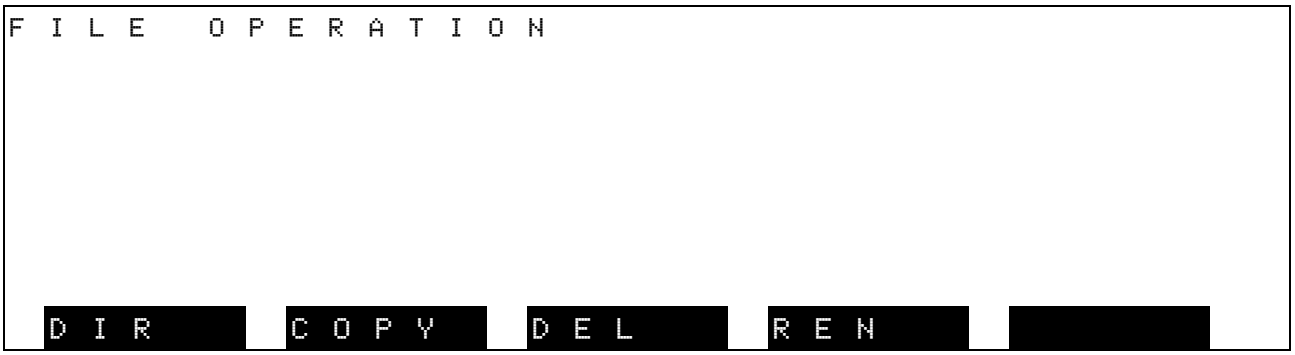
> *MEMO*
> *When the TP1000 is used, data cannot be backed up to an SD card in a batch process. Follow the procedure below for each required file.*

1 Press the NEXT key until "FILE" is displayed in the function menu on the main screen.



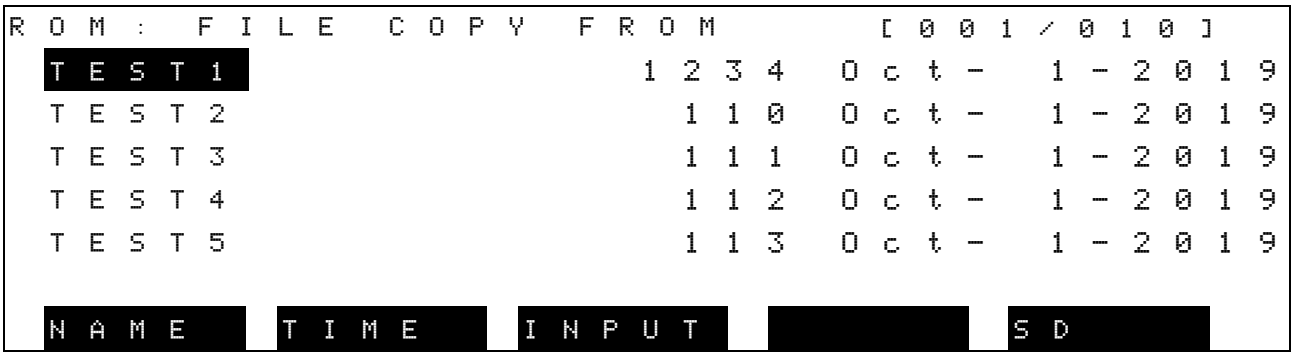
2 Press the FILE (F5) key.

The file operation screen is displayed.



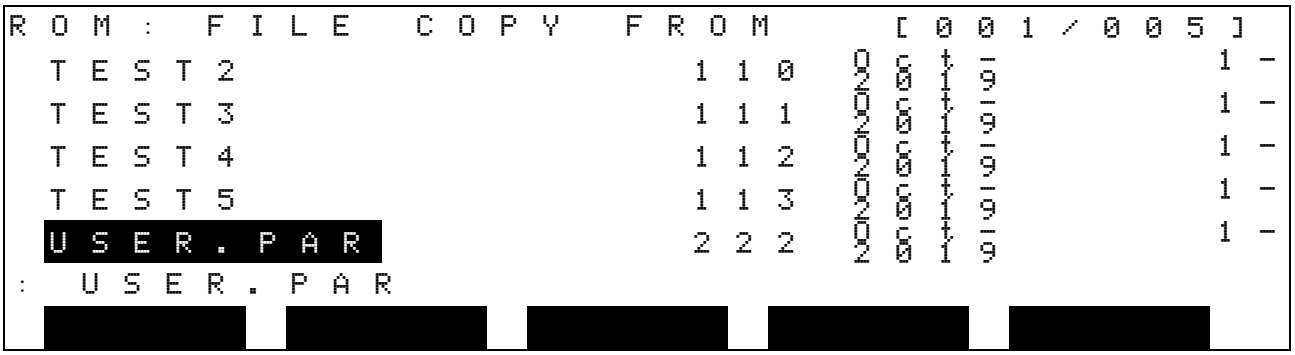
3 Press the COPY (F2) key on the file operation screen.

The operation file selection screen is displayed.



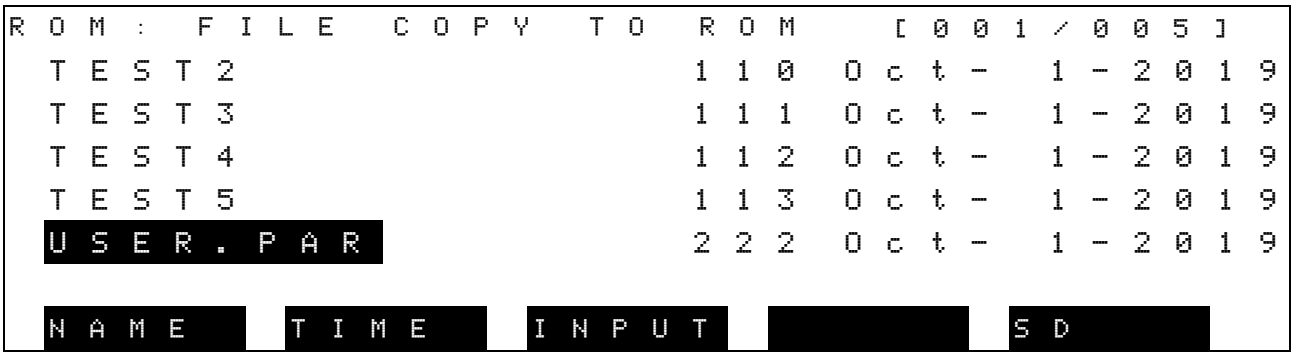
4 Move the cursor to the location of the file you want to copy and press the EXE key.

The name of the file is displayed on the command line.



5 Press the EXE key.

The operation file selection screen is displayed again.

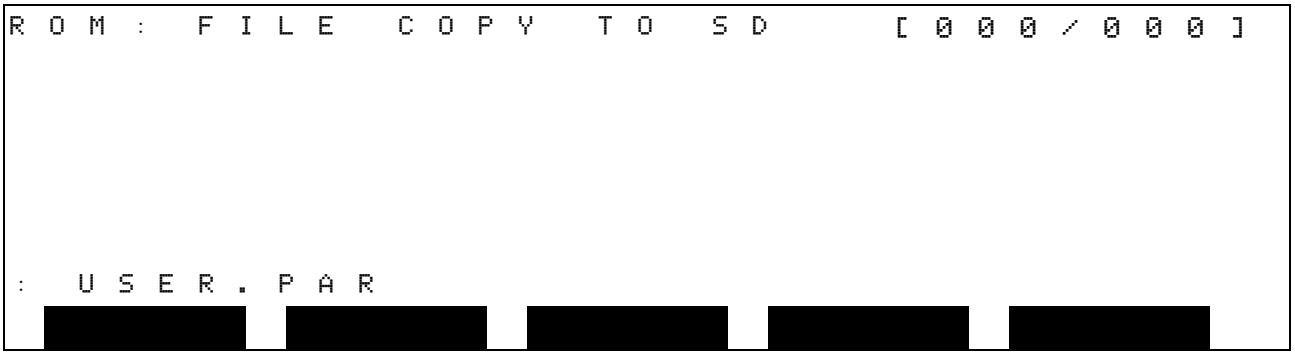


6 Press the SD/ROM (F5) key.

The operation file selection screen of the SD card (ROM) is displayed.

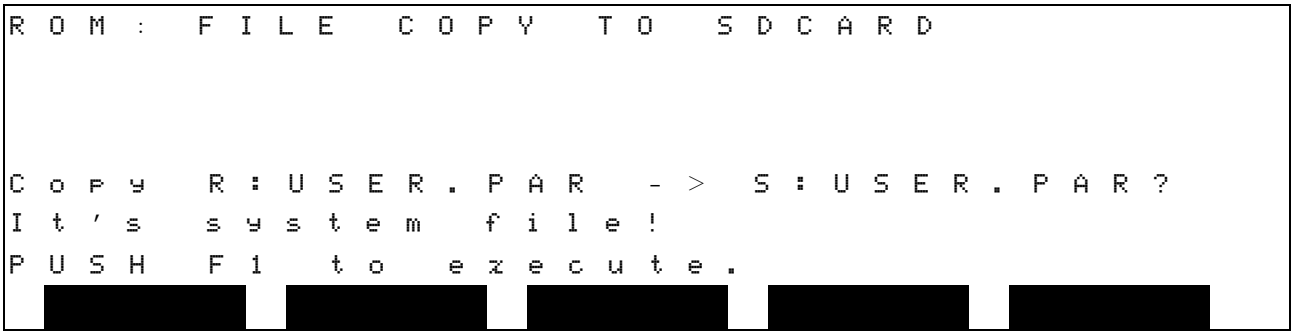
The name of the source file is displayed on the command line.

If the file to be copied is a system file, refer to step 7. Otherwise, refer to step a.



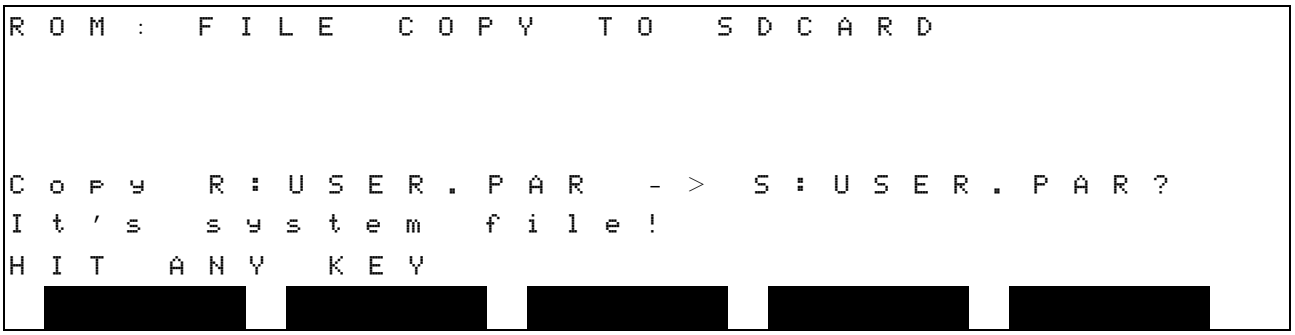
7 Confirm the file name of the copy source file, and press the EXE key.

A confirmation message is displayed.



8 Press the F1 key.

If there is no file with same name as the copy target file, the file copy process is performed.



The following screen is displayed when TEST1 (a program file) is selected before step 6.

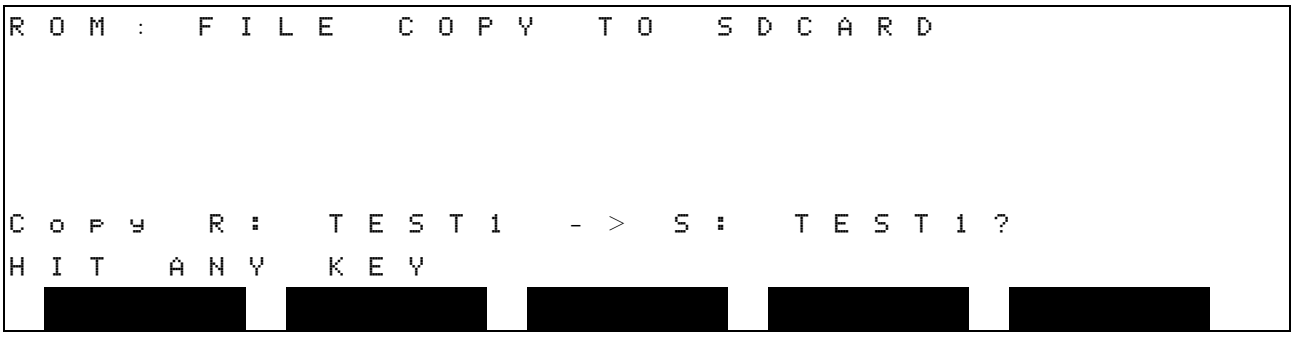
a Confirm the name of the file and press the EXE key.

A confirmation message is displayed.



b Press the EXE key.

If there is no file with same name as the copy target file, the file copy process is performed.



7.4 Restoring Data

This section explains how to restore your data.

7.4.1 Data Restoration (TP5000)

This section explains how to restore the data backed up in 7.3.1.

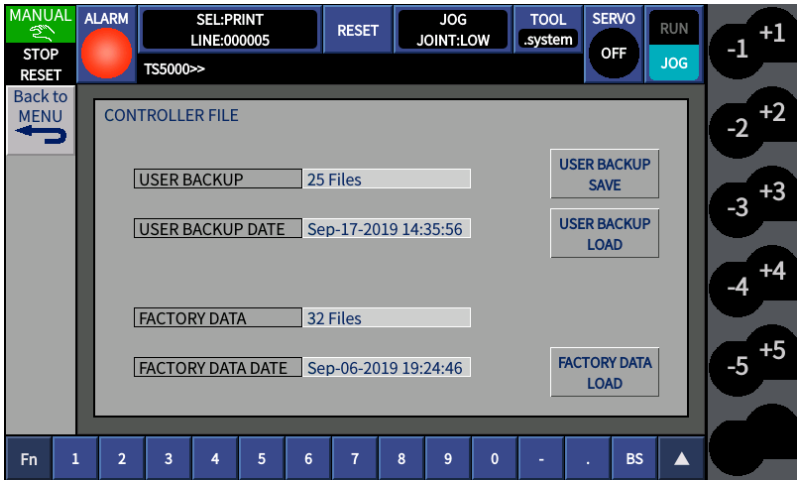
1 After the controller is started, insert the SD card containing the backup files into the TS5000 SD card slot.

2 Switch the user type to “MAINTE”.

For the detailed operating procedure, see "7.2 TP5000 Operation Restrictions".

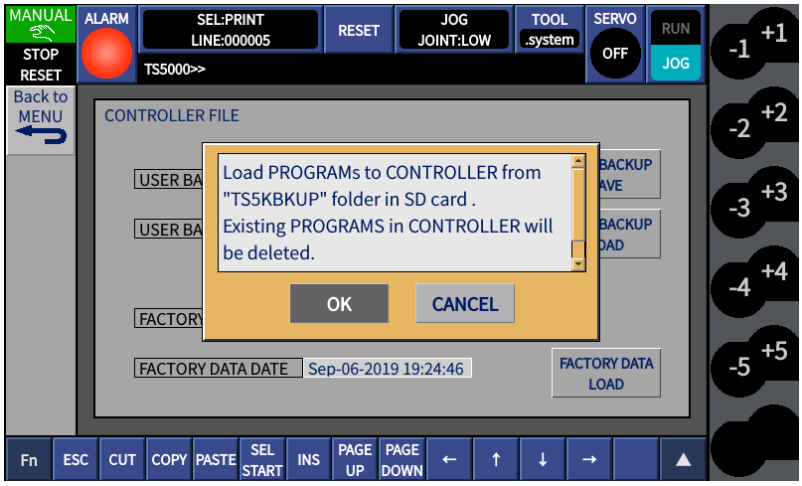
3 Press the “BACKUP” button on the maintenance screen.

The backup screen is displayed.



4 Tap the “Load Backup" button .

A backup loading confirmation window appears.



5 Tap the OK button.

Overwrite the files in controller with the files located in the TS5KBKUP folder on the SD card.

NOTICE

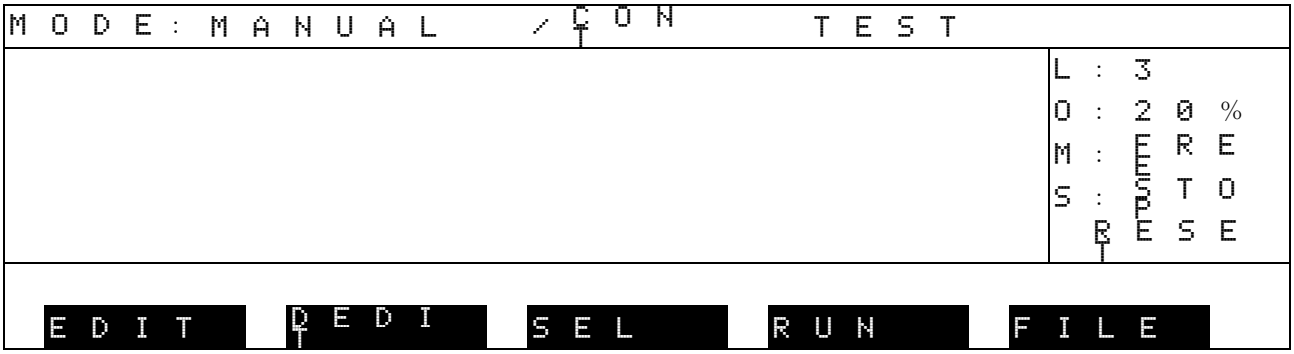
- All files in the controller will be deleted by USER BACKUP LOAD. Before executing, use the file transfer or other function to save the files in the controller to an external device.

7.4.2 Data Restoration (TP1000)

This section explains how to restore the data backed up in 7.3.2.

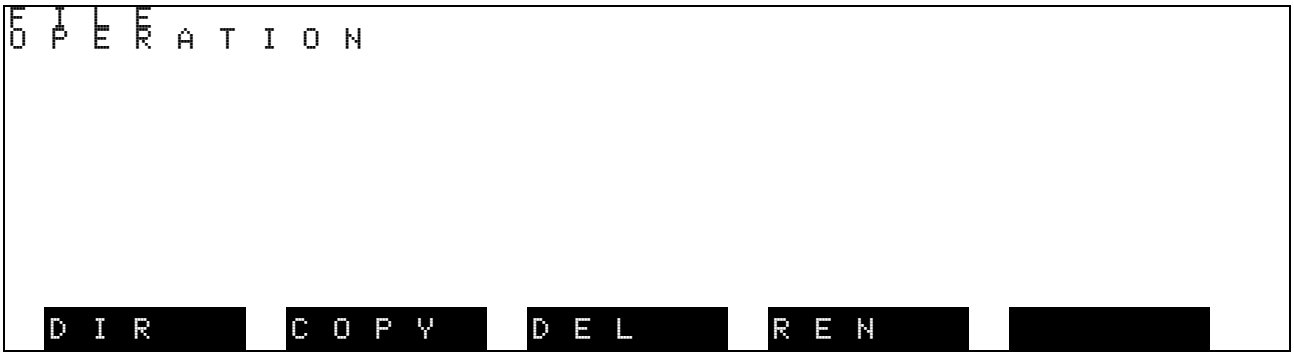
> *MEMO*
> *When the TP1000 is used, data cannot be restored from an SD card in a batch process. Follow the procedure below for each required file.*

1 Press the NEXT key until "FILE" is displayed in the function menu on the main screen.



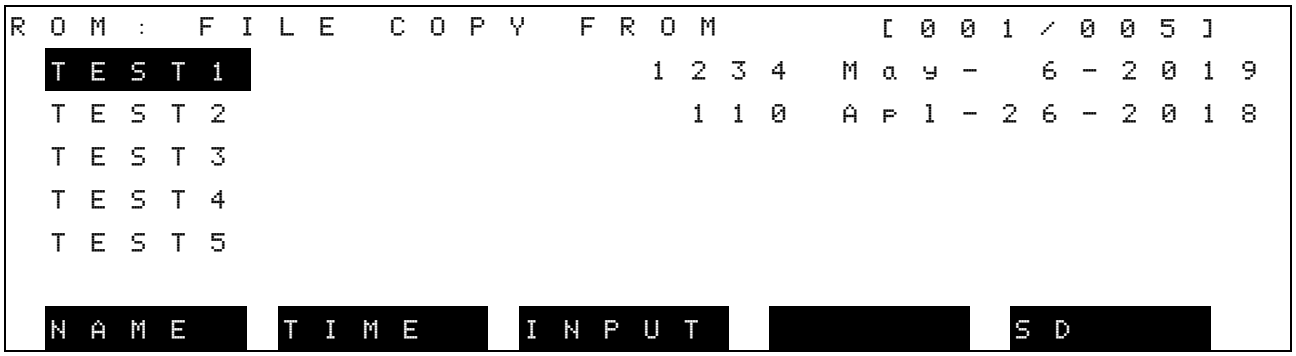
2 Press the FILE (F5) key.

The file operation screen is displayed.



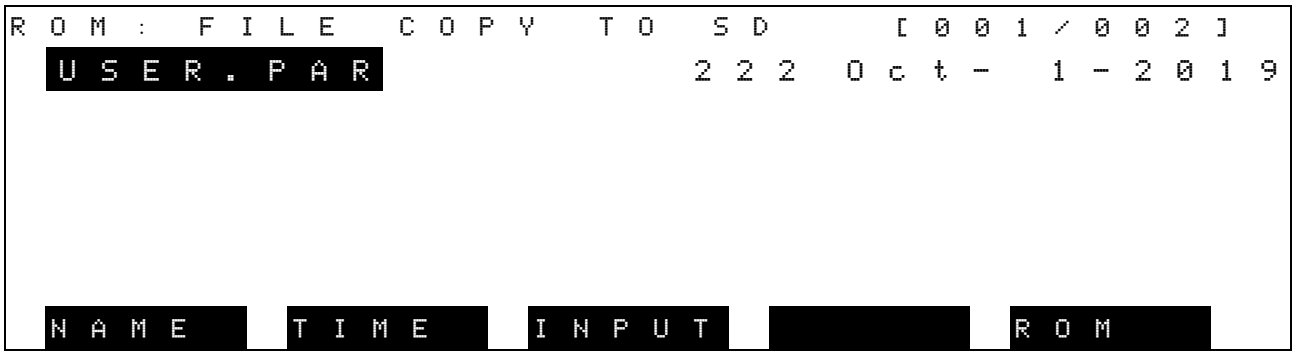
3 Press the COPY (F2) key on the file operation screen.

The operation file selection screen is displayed.



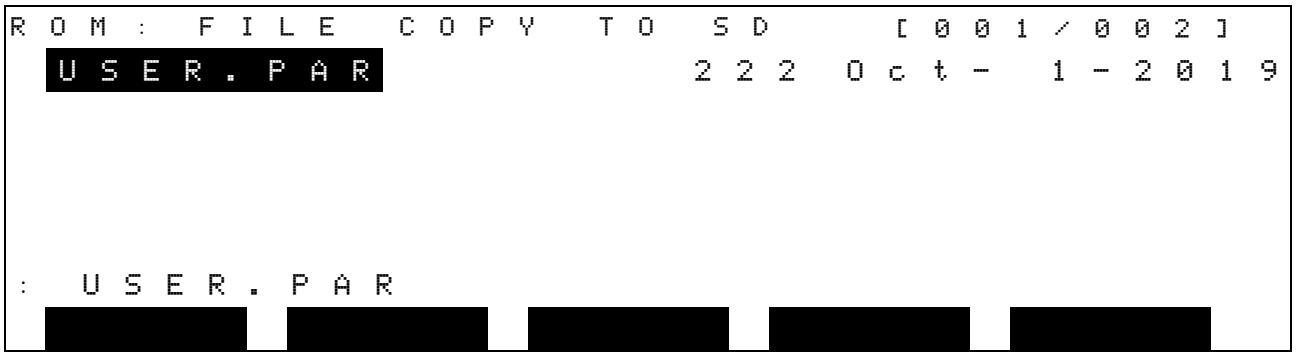
4 Press the SD/ROM (F5) key.

The operation file selection screen of the SD card (ROM) is displayed.



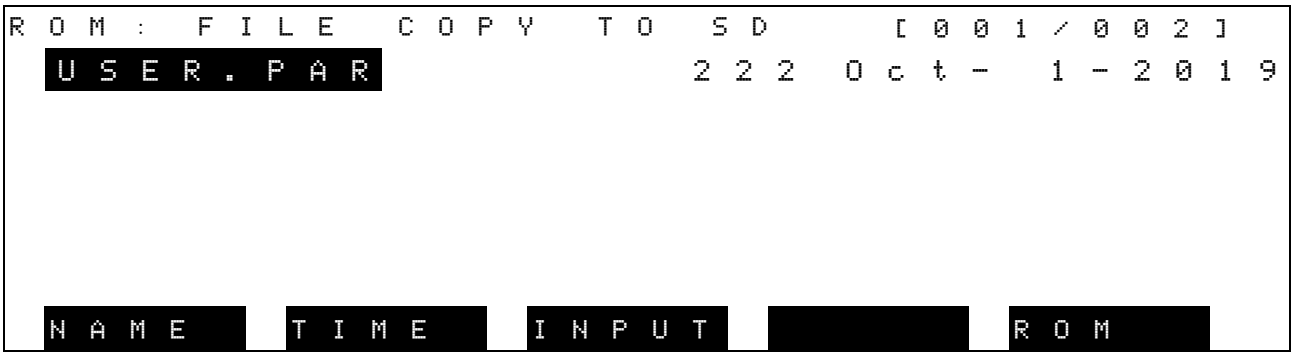
5 Move the cursor to the location of the file you want to copy and press the EXE key.

The name of the file is displayed on the command line.



6 Press the EXE key.

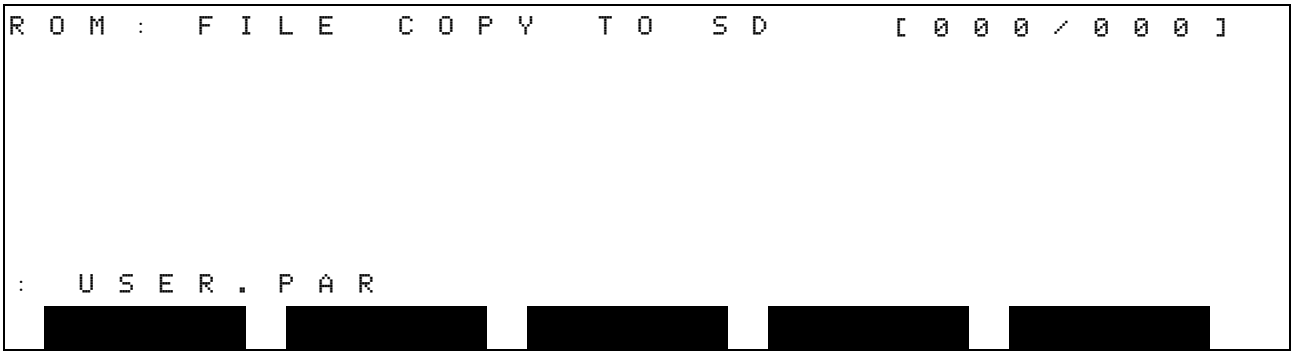
The operation file selection screen is displayed again.



7 Press the SD/ROM (F5) key.

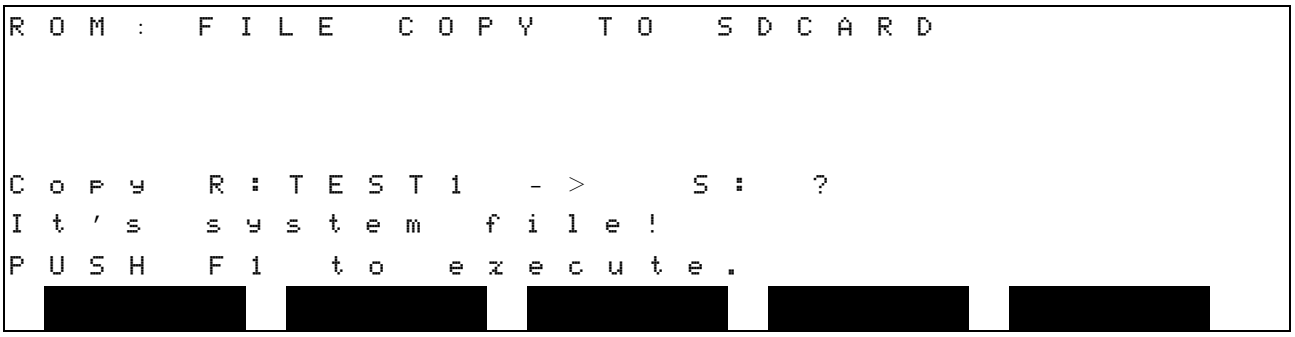
The operation file selection screen of the SD card (ROM) is displayed.

The name of the source file is displayed on the command line.



8 Confirm the file name of the copy source file, and press the EXE key.

A confirmation message is displayed.



9 Press the F1 key.

If there is no file with same name as the copy target file, the file copy process is performed.



7.4.3 Restoring to the Factory Default Settings

This section shows how to restore to the factory default settings.

> *MEMO*
>
> *When this procedure is executed, the controller status returns to the factory default settings. Be sure to back up any necessary parameters, programs, and other data that has been modified by referring to "7.3 Data Backup".*

1 Create a TS5KBOOT folder on SD card and store the FACTBKUP.BIN file in the folder.

TS5KBOOT folder
└ FACTBKUP.BIN

> *MEMO*
>
> *1. Use an SD card formatted in FAT16 or FAT32.*
> *2. FACTBKUP.BIN is stored on the supplied DVD.*

2 Insert the SD card into the controller while the power is turned off.

3 Turn on the power and start the controller.

The following screen is displayed only when the TP1000 is connected.

```
TS5000  RSW:0  System Update
X8KCB-01B  2019-10-01  00:00  AAAA
S0688B     2019-10-01  00:00  AAAA
X8KSB-01B  2019-10-01  00:00  AAAA
X8KCRB-01B 2019-10-01  00:00  AAAA
X8KCRX-01B 2019-10-01  00:00  AAAA
MAC  ADDR  :  02.00.00.00.00.01
   Press the push switch to start
```

4 After startup, if there is repetitive sound alarm, press and hold the maintenance switch for about 3 seconds.

The alarm sound will turn off because the file is being written.

5 After writing of the BIN file is completed, if there is repetitive sound alarm, turn off the power.

Restoration to the factory default settings is complete.

7.5 Maintenance Procedure

7.5.1 Checking the vent holes of the controller

If the vent holes are blocked, the temperature inside the controller will rise, resulting in malfunction. Check the following to ensure that the vent holes are not blocked.

- Remove any objects in front of the vent holes to ensure adequate ventilation.
- Remove any objects stuck in the vent holes.

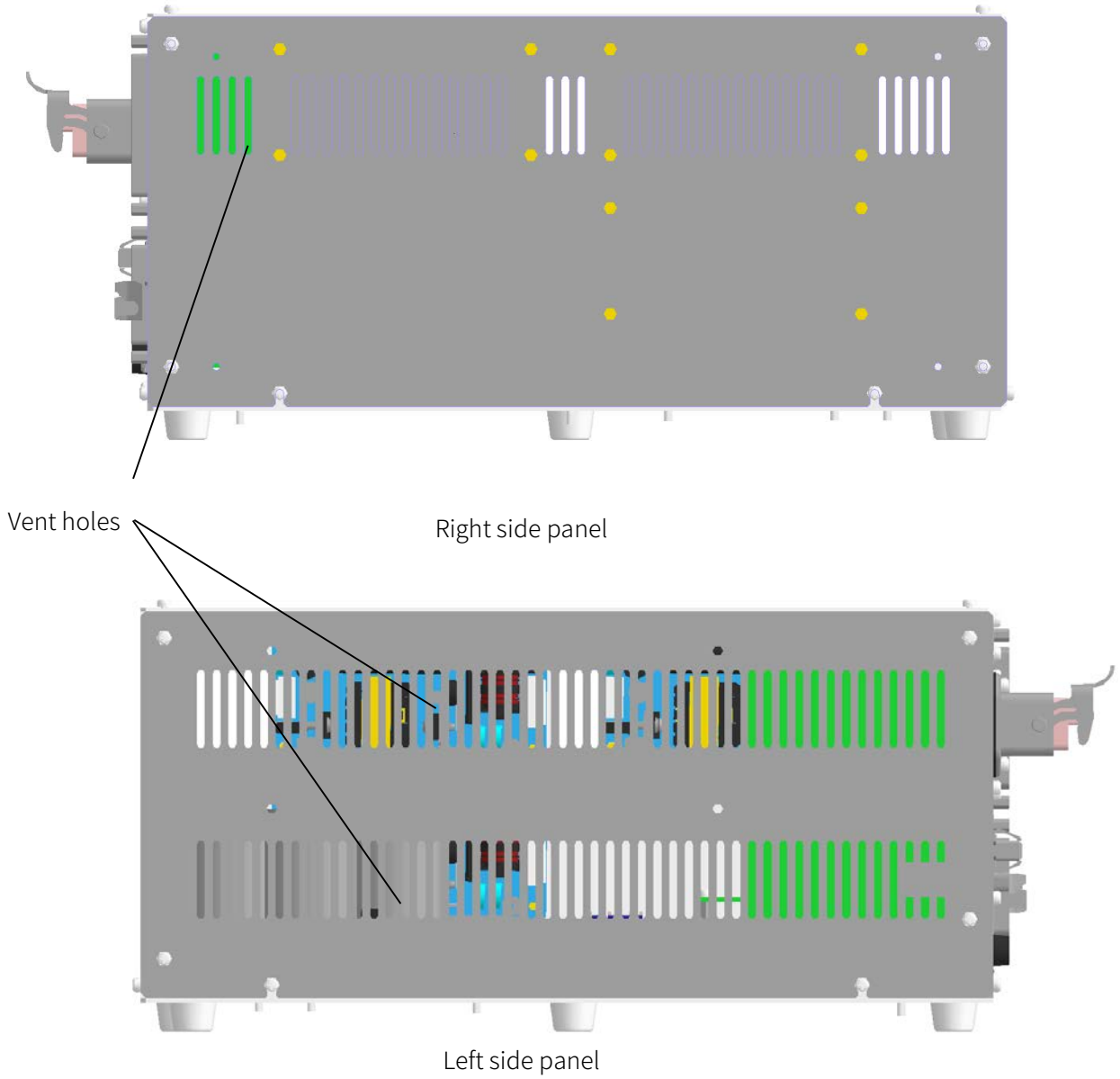


Figure 7.1 Controller Side Views

7.5.2 Safety device operation check (TP5000)

Check that the emergency stop switch and other switches of the teach pendant and safety devices connected to the external operation input signals function normally.

Confirm that the emergency stop switches on the teach pendant work normally.

1. **Turn the master mode switch ① on the front of the TP5000 to the MANUAL position.**
2. **Turn the emergency stop switch ② clockwise to confirm that the emergency stop is reset.**
3. **While holding the Enable switch ③ of the teach pendant at the neutral position, press the "Servo ON" button on the teach pendant screen to turn on the servo. Confirm that the "SVON" LED on the front of the controller has been turned on and that the screen has been changed.**
4. **Press the emergency stop switch ② on the teach pendant, and confirm that the servo is turned off. Confirm that the "SVON" LED on the controller front has been turned off and that the screen has been changed.**
   * In this state, check that the emergency stop switch ② is in the pressed-down state.
5. **Turn the emergency stop switch ② clockwise and confirm that it is reset.**

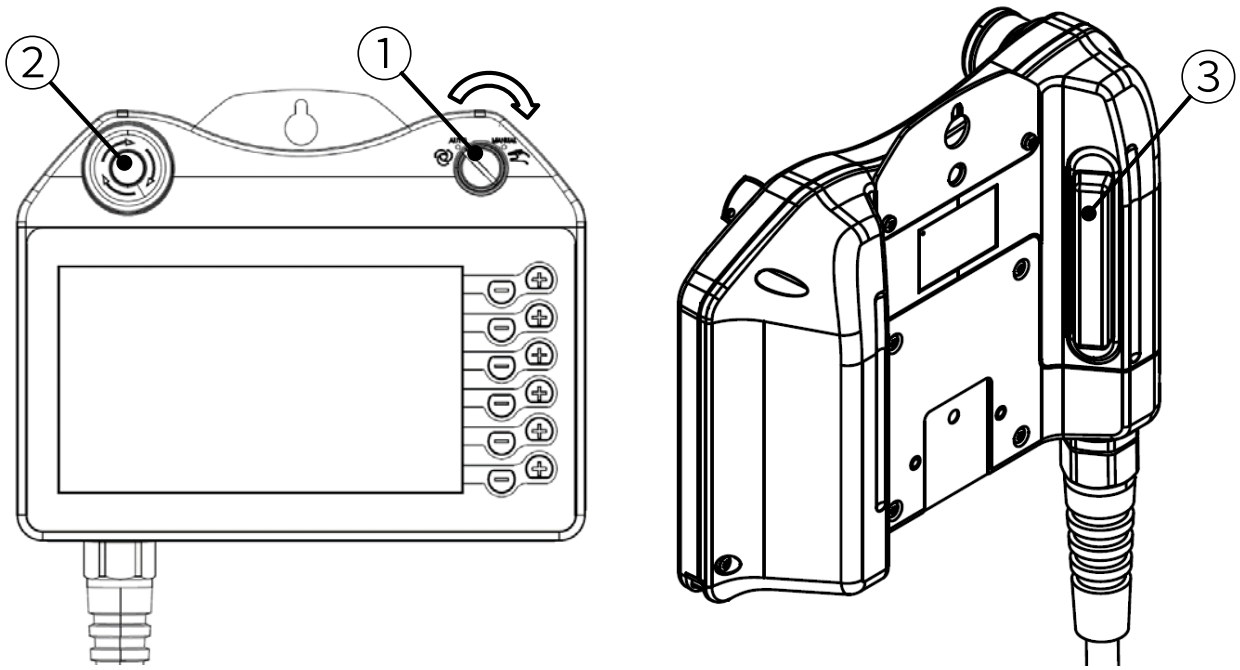


Figure 7.2 Teach Pendant Emergency Stop Switch (TP5000)

Check that the external power supply for input/output and that the safety devices connected to the EMGSTOP connector are functioning normally.

1 **Confirm that the servo will be turned off by turning on the servo for the robot and then turning on the external emergency stop switch connected to the EMGSTOP connector.**

7.5.3 Safety device operation check (TP1000)

Confirm that the emergency stop switch and other switches of the teach pendant and safety devices connected to the external operation input signals function normally.

Confirm that the emergency stop switches on the teach pendant work normally.

1. **While in MANUAL mode, turn the emergency stop switch ① clockwise to confirm that the emergency stop switch ① is turned off.**
2. **While holding the Enable switch ③ of the teach pendant at the neutral position, press the Servo ON switch ② on the teach pendant to turn on the servo. Confirm that the LED of the Servo ON switch ② is turned on.**
3. **Press the emergency stop switch ① of the teach pendant and confirm that the servo is turned off. The LED of the Servo ON switch ② turns off.**
4. *** In this case, confirm that the emergency stop switch ① is in the pressed state.**
5. **Turn the emergency stop switch ① clockwise and confirm that it turns off.**

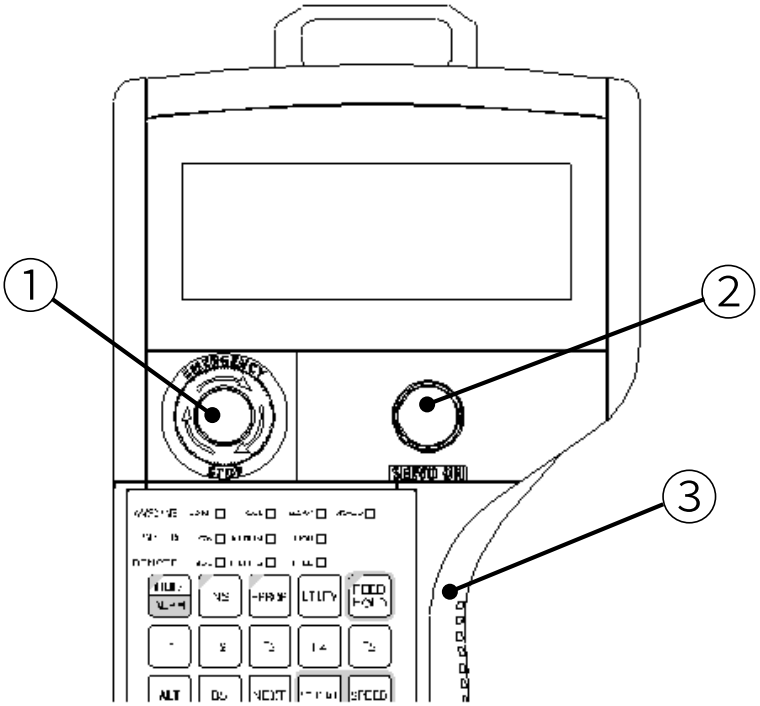


Figure 7.3 Teach Pendant Emergency Stop Switch (TP1000)

Check that the safety device connected to the EMGSTOP connector is functioning normally.

1. **Confirm that the servo will be turned off by turning on the servo for the robot and then turning on the external emergency stop switch connected to the EMGSTOP connector.**

7.5.4 Replacing the controller battery

Controller maintenance and inspection must be performed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country. The Shibaura Machine Service Department will perform replacement work. A fee is charged for this service. Be aware that Shibaura Machine's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

⚠DANGER

- Do not incinerate, disassemble, short-circuit, or charge the batteries. The batteries could rupture, resulting in a fire. Dispose of the battery in compliance with the customer's regulations.

The TS5000-SS uses button cell batteries for running the real-time clock (RTC).
The controller battery should be replaced about every 5 years. After the button cell battery is used for a certain period of time, the battery voltage will drop, the date and time information will be lost, and in certain cases, battery fluid could leak, resulting in a breakdown. After replacing the battery, set the date and time.
Also, do not remove the button cell battery except when replacing it. Otherwise, the date and time data saved internally will be lost.
Replace the battery if any of the following alarms occurs.
"MAIN Battery alarm" (Alarm code 001-145)

Table 7.1 Recommended Battery

Button cell battery type	Manufacturer	Battery life
CR2032H	Maxell Holdings, Ltd.	Follow the manufacturer's recommendations.

Replacement Procedure

1 **Turn off the main circuit breaker for the power supply, and check that the controller is turned off.**

2 **Remove the power cable from the ACIN connector.**

3 **Obtain a new button cell battery, remove the SEMS screws (M3×8×6) securing the cover, and remove the cover from the controller body. (*After replacing the batteries, remount the cover so that there is no looseness or rattling in the screws.)**

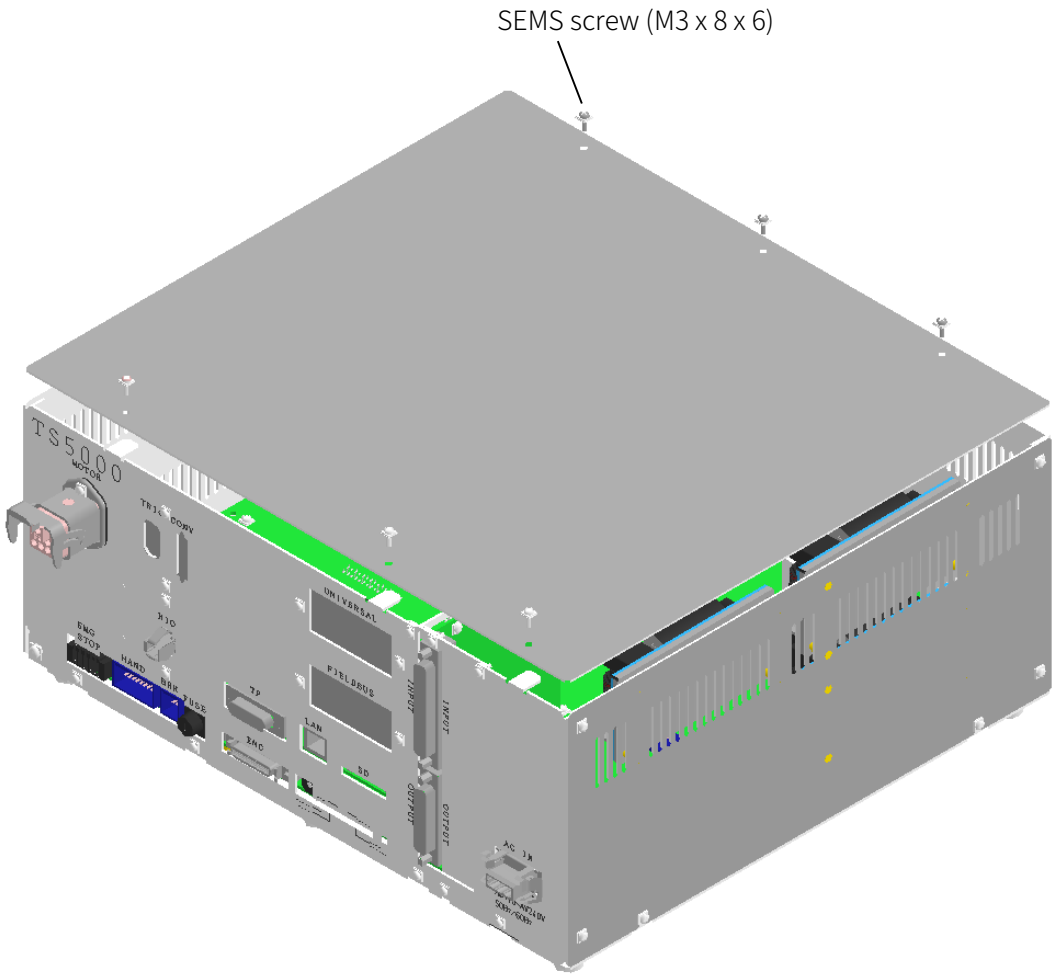


Figure 7.5 Removing the Cover

4 **To remove the battery, press the button cell battery in the direction of the arrow ① as shown in Figure 7.6, and remove by lifting up in the direction of the arrow ② while being careful not to get caught in the battery holder clips.**

*** To insert the button cell battery, perform the above procedure in reverse.**

Even if the button cell battery was not used, its power is reduced due to natural discharge over time. Use a button cell battery that was purchased within the last three years (stored at room temperature).

5 **After installing the button cell battery, use the SEMS screws (M3×8×6) to remount the cover to the controller body.**

6 **Set the date and time after the controller is started for the first time after battery replacement.**

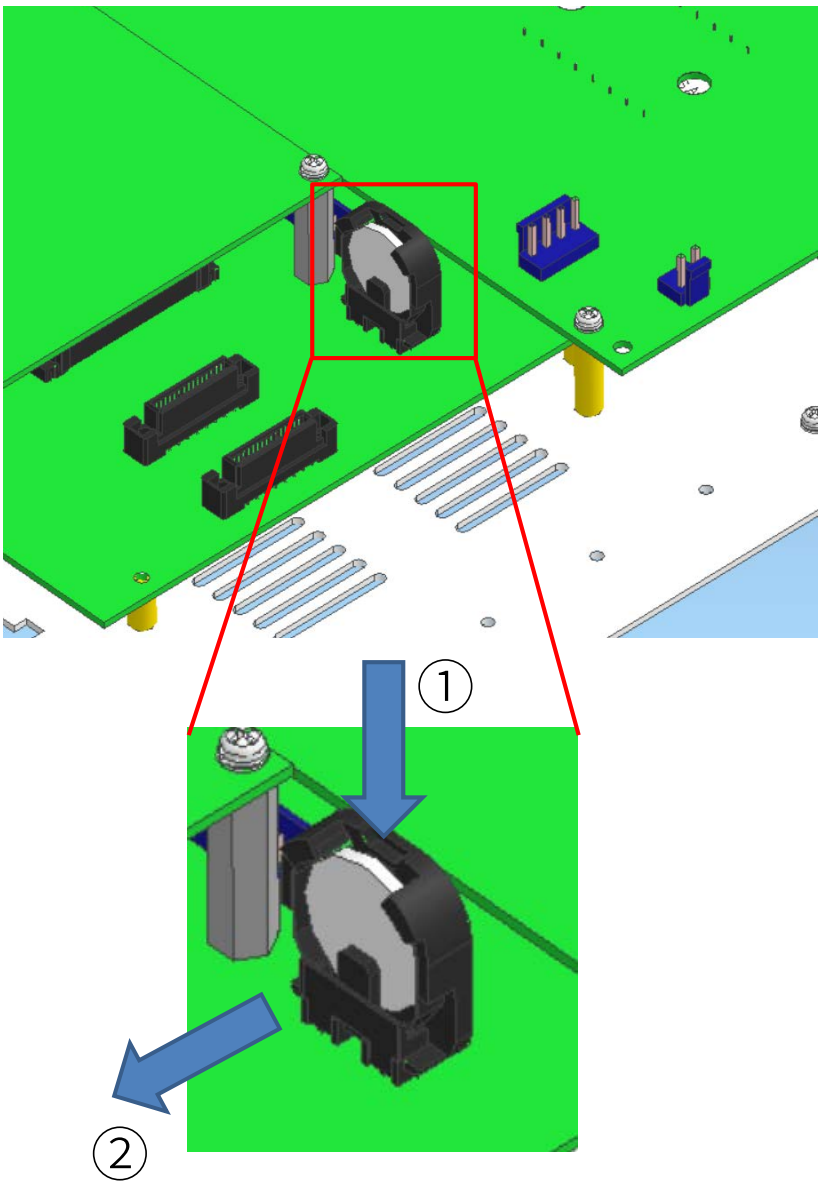


Figure 7.6 Removing the Button Cell Battery

7.5.5 Replacing the switching power supply

Controller maintenance and inspection must be performed by qualified personnel. These processes must be compliant with the laws, regulations, and ordinances of the respective country. The Shibaura Machine Service Department will perform replacement work. A fee is charged for this service. Be aware that Shibaura Machine's warranty does not cover failures and accidents resulting from the replacement of the parts by the customer.

⚠DANGER

- Install the switching power supply as described in the instruction manual. Installing the unit upside-down or with the right and left sides reversed, or using a different connector could cause a short-circuit or disconnection, which could result in a fire. It could also result in electric shock due to electric leakage or other causes.

The lifespan of the switching power supplies used for the controller depends on the usage conditions. The lifespan of the power supply may vary from 5 years to 10 years depending on the usage conditions, but we recommend replacing the power supply every five years as preventive maintenance.

Replacement Procedure

1. **Turn off the main circuit breaker for the power supply, and check that the controller is turned off.**
2. **Remove the power cable from the ACIN connector.**
3. **Use a #2 Phillips screwdriver to remove the SEMS screws (M3×8) that are securing the cover, and remove the cover from the controller body.**

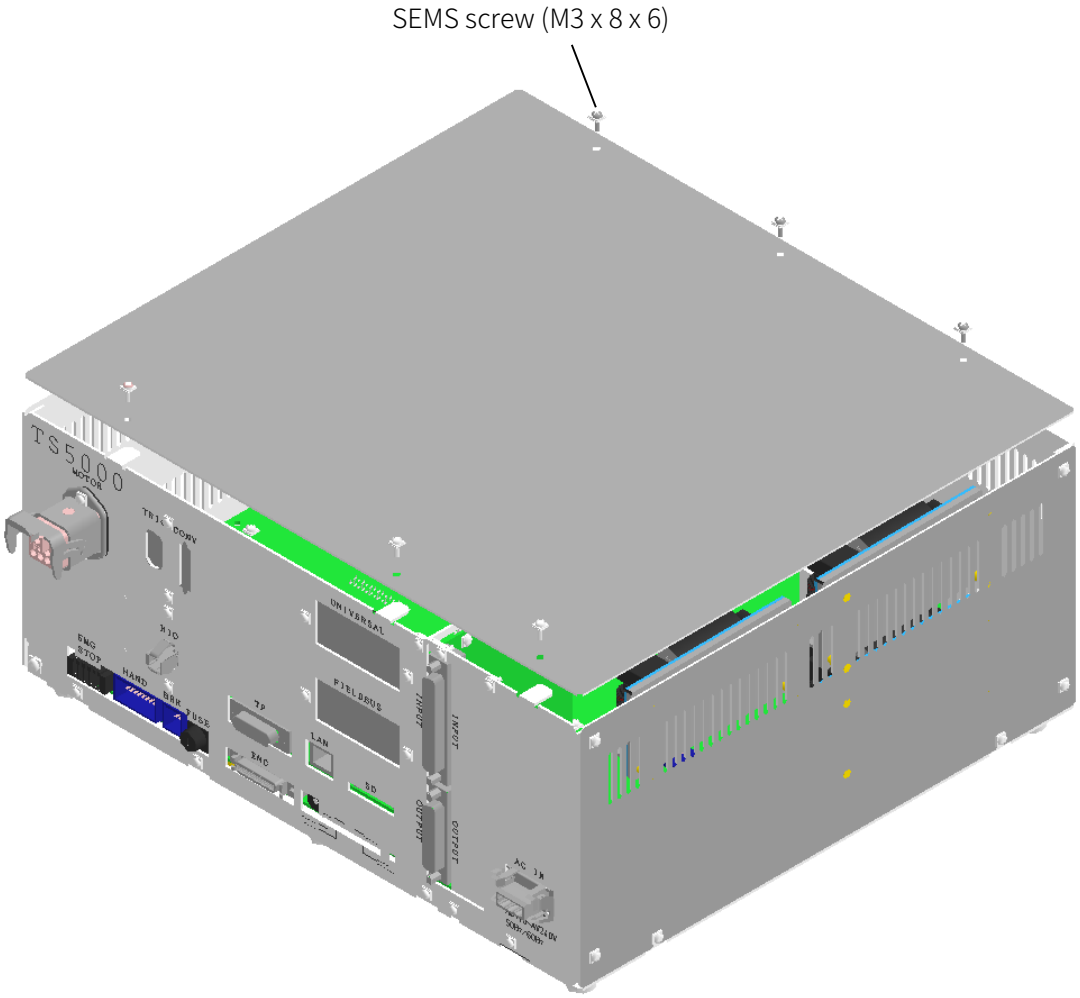


Figure 7.7 Removing the Controller Cover

4 **Remove the cables (switching power supply primary and secondary sides) connected to the switching power supplies.**

5 **Use a Phillips #2 screwdriver to remove the SEMS screws (M3×8), and replace the three switching power supplies.**

6 **Use the SEMS screws(M3×8) to mount the new switching power supplies, then reconnect the cables (switching power supply primary and secondary sides).**

7 **Use the SEMS screws (M3×8×6) to remount the cover to the controller's body. (*After replacing the switching power supply, remount the cover so that there is no looseness or rattling of the screws.)**

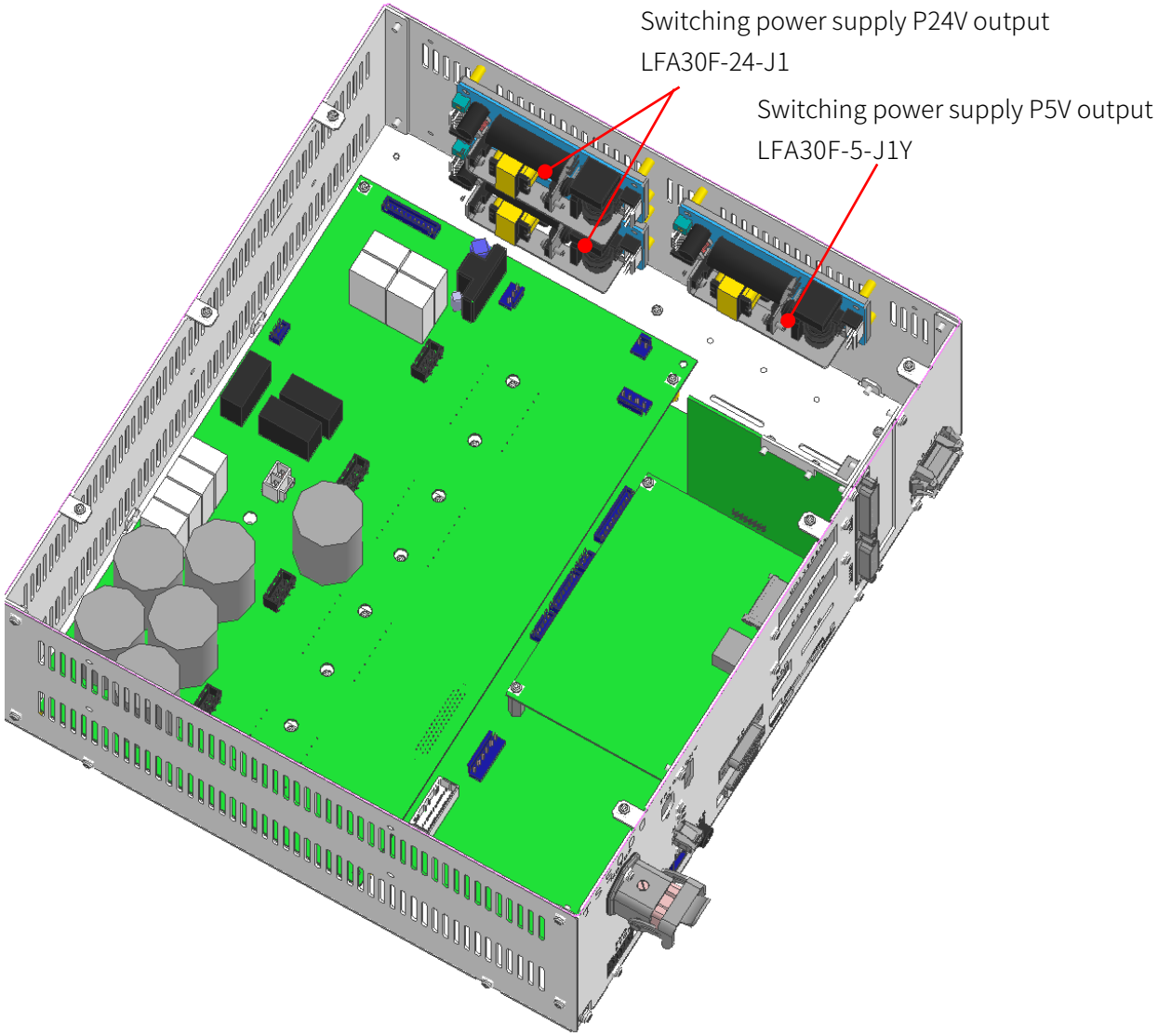


Figure 7.8 Top View of Inside of Controller

7.5.6 Fuse replacement

The fuse installed on the front of the controller will blows when a current higher than the specification value flows in the I/O section.
If the "I/O Fuse Broken" (8-273) alarm is displayed, the fuse has blown. Replace the fuse.
If the fuse of the output module blows out, thoroughly investigate which circuit failed, and remove the cause before replacing the fuse.

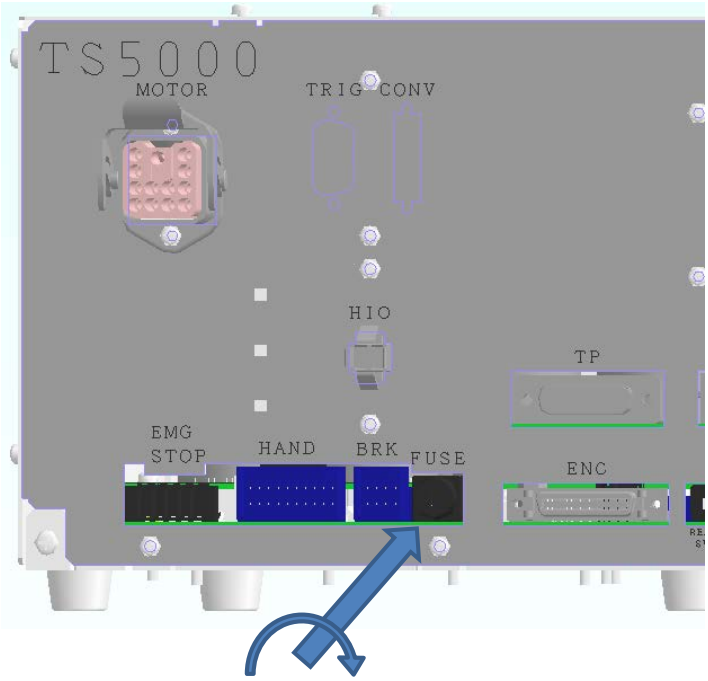


Figure 7.9 Controller Front View

Table 7.10 Fuse Replacement Table

Fuse type	Manufacturer
51NM030H	Pico Inc.

Replacement Procedure

1. **Turn off the P24V power supply connected to the controller and EMGSTOP connector.**
2. **Use a #2 flat head screwdriver to remove the fuse holder in the figure above. (Turn 90° counter-clockwise while pressing down to unlock.)**
3. **Remove the fuse, and attach a new fuse to the fuse holder.**
4. **Mount the fuse holder. (Turn 90° clockwise while pressing down to lock.)**
5. **Turn on the P24V power supply connected to the controller and EMGSTOP connector, and confirm that no error occurs.**

8. Replacement Parts for Maintenance

8.1 Precautions Regarding Replacement Parts

⚠ WARNING

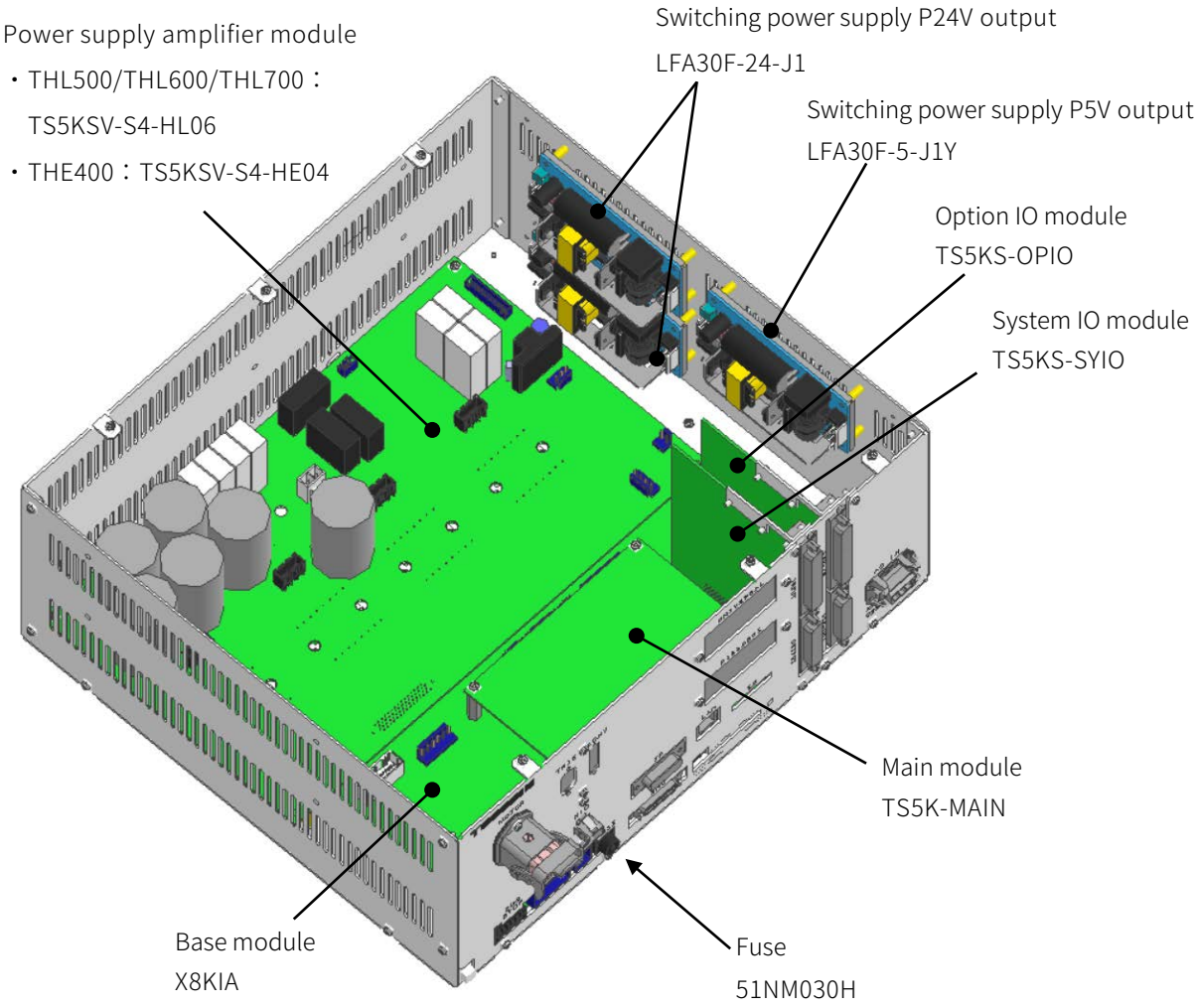
- Be sure to replace parts with the Shibaura Machine-specified service replacement parts. Use of unauthorized parts could cause a fire.
- Except when replacing fuses, be sure to always request replacement work from the Shibaura Machine servicing company. A fee will be charged for this work. Shibaura Machine will not be liable for any accidents or damage that arises due to replacement work performed by the customer or its results.

NOTICE

- Be sure to replace parts with the Shibaura Machine-specified service replacement parts. Use of unauthorized parts could lead to reduced robot performance and lifespan. This could also lead to a breakdown.

8.2 Replacement Parts List

No.	Product name	Model	Drawing No.	Manufacturer	Qt'y used	Notes
1	Switching power supply	LFA30F-5-J1Y	T016684	COSEL CO., LTD.	1	P5V power supply
2	Switching power supply	LFA30F-24-J1	T016685	COSEL CO., LTD.	2	P24V power supply
3	Button cell battery	CR2032	T016686	Maxell Holdings, Ltd.	1	
4	Fuse	51NM030H	T016687	Pico Inc.	1	
5	Power supply amplifier module (THL500/THL600/THL700)	TS5KSV-S4-HL06	L33834G02	Shibaura Machine Co., Ltd.	1	
	Power supply amplifier module (THE400)	TS5KSV-S4-HE04	L33834G01		1	
6	System IO module	TS5KS-SYIO	L27670G03		1	
7	Option IO module	TS5KS-OPIO	L27670G04		1	
8	Main module	TS5K-MAIN	L27669G01		1	
9	Base module	X8KIA	L33805G01			
10	Teach pendant	TP5000	L27664G01		1	5m specifications


Power supply amplifier module
・THL500/THL600/THL700：
TS5KSV-S4-HL06
・THE400：TS5KSV-S4-HE04
Switching power supply P24V output
LFA30F-24-J1
Switching power supply P5V output
LFA30F-5-J1Y
Option IO module
TS5KS-OPIO
System IO module
TS5KS-SYIO
Main module
TS5K-MAIN
Base module
X8KIA
Fuse
51NM030H

8.3 Accessories List

The entire set of accessories can be purchased with the Drawing No. L33824G01.

No.	Product name	Model	Manufacturer	Qt'y used	Notes
1	I/O jumper connector	M067817	Shibaura Machine Co., Ltd.	1	For INPUT
2	TP5000 jumper connector	S981572	Shibaura Machine Co., Ltd.	1	For TP
3	Connector	XM2S-2511	Omron	1	For OUTPUT shell
4		XM3A-2521	Omron	1	For OUTPUT plug
5		231-103/037-000	WAGO Kontakttechnik GmbH & Co. KG	1	For ACIN
6		B2CF3.50/10/180 LR	Weidmüller Interface GmbH & Co. KG	1	For EMGSTOP
7	SAFE jumper cable	S981311	Shibaura Machine Co., Ltd.	2	For EMG jumper
8	Fuse	51NM030H	Pico Inc.	1	External 24 V fuse

Disposal

This chapter describes the disposal of the controller.

9. Disposal

Dispose of this product in compliance with the customer's regulations. Shibaura Machine does not provide collection or disposal services.

⚠ WARNING

- Never attempt to incinerate, short-circuit, charge, disassemble, or heat the batteries. They could rupture and leak, and if the battery fluid gets on your skin or eyes, it will cause inflammation.
  - If battery fluid gets on your skin, immediately flush with water to wash off.
  - If battery fluid gets in your eyes, immediately flush with water to wash off, and then seek medical attention.
  - Dispose of the battery in accordance with the customer's regulations.
- Turn off the power before connecting and disconnecting cables from the controller. Touching the terminals of any cables that were disconnected while power was flowing could result in an electric shock.
- Wait about two minutes after disconnecting the power supply before touching the terminals of the controller. Residual voltage may remain in the high-capacity capacitors in the controller, which could result in an electric shock.

⚠ CAUTION

- Wear protective gloves and shoes when carrying the controller by hand. Dropping could result in crushed hands or feet.
- Use a cart or similar device for transporting and carrying. If carried by hand, the controller could be dropped, resulting in crushed hands or feet.

NOTICE

- Dispose of any unneeded packaging materials in compliance with the customer's regulations. When storing the packing materials, be sure to manage them so that they are not used for purposes other than packing the robot.